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USING SENTIMENT ANALYSIS TO ANALYSE COMPANY REPUTATION

ABSTRACT

Reputation analysis can assist businesses in understanding consumer and public perception, which helps them attract investors, build customer trust, and analyse potential dangers that may be encountered. **Sentiment analysis** is one way to achieve this, and the sentiments and keywords can help a company in strategic planning, enhance communication, and create a compelling brand image in the marketplace.

This study examines how social media affects a company's reputation by focusing on Volkswagen and the December 2024 manufacturing shutdown news. Reddit is used as the primary data source for sentiment analysis, utilizing the VADER (Valence Aware Dictionary and sEntiment Reasoner) model to assess the content and comments under relevant posts discussing the topic. The data collection period spans from July to December 2024 and is structured across different event phases.

To evaluate how this sentiment may affect corporate reputation, Volkswagen’s stock price is used as the proxy metric. Stock price levels can influence the degree of informed trading and, consequently, the informativeness of prices about a firm's future performance (Chan et al., 2017).

Later, the study focuses on correlating the sentiment scores derived from the initial analysis with Volkswagen's (VW) stock prices using Yahoo Finance data, in order to examine the influence of these online attitudes in the real world, which can further aid in reputation analysis. Although no significant correlation was found, some visible alignment between sentiment trends and the stock price movement was observed during the speculation period. All the findings are presented using visual tools such as line graphs, pie charts, and scatter plots.

Keywords: Sentiment analysis, Volkswagen, Reddit, corporate reputation, stock price, VADER, factory shutdown, social media analysis

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CHAPTER 1: INTRODUCTION

1.1 Industry and Company Background

Corporate reputation is considered a crucial intangible asset in today's business environment, helping to shape stakeholder perspective and also to influence a firm's financial performance, competitive market position, and long-term sustainability (Gatzert 2015). In the highly competitive automobile sector, a company's reputation is linked to attributes such as safety, innovation, and environmental responsibility. Reputable businesses in this market are well-positioned to attract qualified personnel, raise capital, and establish a strong brand. In 2014, the Reputation Institute ranked firms such as BMW, Toyota, Nissan, Volkswagen, Honda, Michelin, and Suzuki as the most reputable in their respective industries (Sandu, 2015).

One notable incident was when, in October 2024, speculations started to circulate and were later confirmed by an official announcement in December. When Volkswagen, one of the biggest automakers in the world, announced that it was closing its factories, the public's reaction was conflicted and drew significant attention.

The Volkswagen Group was established in 1937, with its headquarters located in Wolfsburg, Germany. It is regarded as one of the biggest automakers in the world, and its well-known brands include SEAT, Bentley, Skoda, Audi, Porche, and Lamborghini. As of April 2025, VW reported a market valuation of $50.344 billion, earnings per share of $24.34, and annual revenue of about $324.66 billion (Britannica 2025). However, the company experienced a 14% drop in net profit in the first half of 2025.

In 2024, VW announced it would close at least three German factories due to declining sales, competition from low-cost Chinese electric vehicles, and high operating costs (DW 2024). Production costs were reported to be 25–50% higher than planned, with 500,000 fewer cars produced annually in Europe than before the pandemic, indicating financial distress (Ziady 2024).

1.1.1 The Shift from Traditional to Social Media-Based Reputation Management

As social media becomes an increasingly important factor, organizations are managing their reputation, engaging stakeholders, and responding to crises using public opinion obtained through these platforms. The traditional one-way method of corporate communication has evolved into a dynamic two-way interaction, where consumers and stakeholders actively influence public opinion and corporate image in real time (Kaul et al., 2016). Brand indices, consumer surveys, and analysis reports are traditional approaches to evaluate corporate reputation, but these often lack the responsiveness required to track real-time shifts in sentiments, especially during important corporate events. They may fail to account for emotional responses, rapid public discourse, and developments that can significantly influence the company’s reputation in this digital age.

This shift requires tools like sentiment analysis to decode user-generated content such as tweets, Reddit threads, and review of the products. The insights obtained from this can help the company identify emerging challenges, understand customer reactions, and adapt or formulate new communication and operational strategies. Lexicon-based models and machine learning approaches are used to understand user-generated content with the intent to extract valuable insights like trends, patterns, and new problems. Sentiment analysis supports organizations in navigating public crises, enhancing customer satisfaction, and redefining marketing strategies. Despite its advantages, sentiment analysis faces challenges such as interpreting sarcasm, handling multilingual content, and addressing ethical concerns related to user privacy and data consent. However, ongoing technological developments in deep learning and Artificial Intelligence (AI) continue to enhance the validity and applicability of sentiment analysis for reputational management (Broklyn et al., 2024).

1.2 Research Purpose and Methodology

This research examines customer sentiment by applying sentiment analysis to Reddit data and investigates the relationship between public sentiment and stock price, which serves as a reputational metric. The analysis aims to provide insights into the impact of corporate decisions on both public perception and financial performance.

This study adopts a quantitative research design using sentiment analysis and correlation analysis. In the initial phase, Reddit comments related to Volkswagen's 2024 factory shutdown announcement were collected and analysed using the VADER (Valence Aware Dictionary and sEntiment Reasoner) sentiment analysis tool. Sentiment scores were then averaged on a weekly basis and compared with the company's stock price data, sourced from Yahoo Finance, where stock price serves as a proxy for corporate reputation.

In the subsequent phase, correlation analysis assessed the relationship between public sentiment and Volkswagen's stock performance. Sentiment trends and key findings were visualized to facilitate interpretation and analysis.

1.3 Research Gap and Contributions

While the relationship between corporate reputation and financial performance is well-documented, limited research addresses real-time stakeholder responses to sudden corporate decisions in the automotive sector. Most existing studies utilize aggregated reputation indices or historical data, rather than analyzing public sentiment related to specific events. Although sentiment analysis is increasingly utilized, its application to a single, high-impact corporate event in conjunction with a reputational metric such as stock price remains insufficiently explored. This study addresses this gap by analyzing public sentiment regarding Volkswagen's planned factory closures in 2024 and evaluating the potential impact on the company's financial reputation.

Contributions

(a) Understanding Sentiment Dynamics

This research provides insight into how real-time customer sentiment responds to significant corporate events, using Volkswagen's 2024 factory closure announcement as a case study.

(b) Linking Sentiment to Reputation

It demonstrates how customer sentiment expressed on social media platforms like Reddit can influence public perception of a company's reputation, particularly when analysed alongside stock price movements.

(c) Data driven Visual Insights

The study delivers valuable visualizations of sentiment trends, aiding in the interpretation of stakeholder reactions and uncovering patterns that can inform strategic decision-making.

1.4 Research Objectives and Questions

Objectives

1. To analyse customer sentiment surrounding Volkswagen's 2024 factory closure announcement events using Reddit data.
2. To evaluate the impact of customer sentiment on social media on company's reputation.
3. To visualize sentiment trends using appropriate data visualization techniques.
4. To correlate the sentiment analysis with company's stock price as an indicator of corporate reputation

Research Questions

RQ 1: How does customer sentiment change during key events for the company?

RQ 2: Can social media sentiment influence company's reputation, as reflected by stock prices?

RQ 3: What visual insights can be derived from the analysis of sentiment trends?

Research Hypothesis

H_0: There is no significant relationship between customer sentiment on social media and Volkswagen's stock price.

H_1: There is a significant relationship between customer sentiment on social media and Volkswagen's stock price.

1.5 Scope and Structure of the Thesis

This study focuses on the automotive industry, where corporate reputation plays a critical role in customer trust and brand equity. This case provides a relevant opportunity to explore how sudden corporate decisions like the one chosen can affect stakeholder sentiment and perceived corporate value as well. The focus of this research is limited to the period surrounding the factory shutdown announcement to capture the immediate customer response and market impact. Considering additional data from similar scenarios that have occurred for the company in the past could also provide useful insights.

Structure Overview

Chapter 1: Introduction

Introduces the research background, explains the importance of corporate reputation and real-time customer sentiment, states the research aim, objectives, and research questions, and outlines the overall thesis structure.

Chapter 2: Literature Review

Identifies research gaps while presenting a summary of the existing literature papers on corporate reputation, social media sentiment analysis, the effect of corporate events on reputation, and sentiment analysis tools.

Chapter 3: Research Methodology

Details the research methodology, including quantitative research design, including the process of data collection from Reddit and stock data using yfinance. It further describes the application of VADER sentiment analysis and correlation analysis methods.

Chapter 4: Results and Analysis

Utilizes data visualizations like charts and graphs to report and analyze the results, including sentiment analysis findings, stock price trends, and the relationship between sentiment and stock price and to gain insights.

Chapter 5: Conclusion and Recommendations

Outlines the main findings, discusses their importance, and proposes suggestion for future research and recommendations for stakeholders in order to conclude the dissertation's findings.

CHAPTER 2: LITERATURE REVIEW

2.1 Sentiment Analysis: Theory and Methodology

Sentiment analysis, also referred to as opinion mining, represents a rapidly developing field within Natural Language Processing (NLP). The paper by Ravi and Ravi (2015) serves as a key reference for studies involving public sentiment from social platforms. It provides a survey that categorizes sentiment analysis tasks based on the document, sentence, and aspect levels, and reviews both machine learning and lexicon-based methods. The main focus was on preprocessing strategies, domain specific challenges, and tools such as Twitter4J and the Facebook Graph Application Programming Interface (API).

Medhat et al. (2014) expanded on this by comparing machine learning, lexicon-based, and hybrid methods, emphasizing that hybrid designs in combination with lexicons and learning-based models are increasingly adopted as they integrate linguistic precision with contextual learning. From a cognitive perspective, Cambria et al. (2017) introduced SenticNet 5 and the concept of affective computing. Their approach uses common-sense knowledge representation and semantic networks to capture the nuanced expressions such as anticipation and surprise, which are often present in complex corporate discussions on social media. This concept aligns with the shift from a surface-level sentiment scoring to a deeper emotional interpretation that enriches reputation-based studies about companies.

There was another update to this framework by Pei et al. (2019) who introduced a taxonomy for Targeted Sentiment Analysis (TSA), distinguishing between Target-Grounded, Targeted Non-Aspect Based, and Targeted Aspect Based sentiment analysis. This distinction can support more precise reputation analysis.

Zhang et al. (2018) detailed how pre-trained embeddings and deep architecture improve the contextual representation and performance across sentiment tasks. This bridges traditional lexicon-based systems with the modern neural frameworks. These studies show that sentiment analysis evolved from a rule-based to a data-driven model that can integrate linguistic, psychological, and contextual cases. Shaik et al. (2023) offer a domain-specific application of sentiment analysis on education where techniques like aspect-level classification and Bidirectional Encoder Representations and Transformers (BERT) based transformer models are highlighted. Even though the education sector was their primary focus, sentiment analysis in the automotive industry can also benefit from the techniques discussed in their study, such as annotation strategies.

Liu (2021) offered an extensive overview of opinion mining, defining it as a computational approach that analyses the opinions, sentiments, and emotions expressed by individuals in textual data. This work introduced the distinction between sentiment orientation, which is positive, negative, and neutral, and opinion targets. This helped in the categorization of sentiment into a multi-level format

2.2 Sentiment Analysis Systems: Methods and Tools

Pak and Paroubek (2010) demonstrated an early method for auto-labelling Twitter data using emoticons, thus producing a stable sentiment corpus. Their model mainly relies on Naive Bayes and part of speech tagging, thus illustrating how Twitter can function as a real time dataset for public opinion mining.

Taboada et al. (2011) developed SO-CAL framework, a lexicon-based approach that adjusted polarity according to intensifiers, negations and valence shifts. Frameworks like this proved that syntactic awareness could enhance the accuracy in textual sentiment classification, an insight which was later adopted by tools like VADER. Agarwal et al. (2011) expanded this foundation by developing a Support Vector Machine (SVM) classifier for Twitter data. Their model demonstrated that machine learning methods could surpass pure lexicon-based systems where there is sufficient data available.

Similarly, Hutto and Gilbert (2015) proposed VADER model, which, a rule-driven sentiment tool built for analyzing microtext such as tweets and comments. It combines human-validated lexicons with syntactic rules, thus achieving superior performance compared to many machine learning models in social media contexts. This tool is thus selected as the key tool to perform sentiment analysis in this research.

In the extensive survey of Twitter sentiment methods, Giachanou and Crestani (2016) reviewed over fifty studies and concluded that lexicon-based systems remain highly effective for microtext such as tweets and comments because of their simplicity, language independence and resilience to noisy data.

In the context of election related sentiment analysis, Mishra et al. (2024) compared the performance of VADER, TextBlob, and BERT. They found that VADER is more effective, especially for informal text while BERT underperforms due to lack of robustness in handling abbreviations and the slang commonly seen in Twitter posts and comments.

In a similar comparative study, Tsiourlini et al. (2024) also compare VADER, TextBlob, and Google's Sentiment API for classifying YouTube comments, where they conclude that Google's API offers the best human-aligned labelling. Nonetheless, VADER still performs competitively when combined with supervised learning models like Support Vector Machine (SVM).

2.3 Role of Sentiment Analysis in Consumer Behaviour and Online Retail

Vinodhini and Chandrasekaran (2012) provided an overview of sentiment analysis techniques and discussed their application in consumer analytics and e-commerce. Their work demonstrates how businesses can gain insights from the contents that are generated online. They reinforced that combining machine learning techniques with lexicon-based sentiment scoring improves accuracy in classification and allows companies to monitor customer preferences and adapt to necessary strategies.

Gräbner et al. (2012) introduced a lexicon-based sentiment analysis system for classifying hotel customer reviews from TripAdvisor. Their study shows that classifying the sentiment into three categories improves the accuracy and reveals the significance of contextual terms, which was mirrored in this research on Reddit comments.

Feldman (2013) specified the connection between sentiment analysis and corporate decision-making by outlining how opinion mining has become important to marketing, brand management, and product evaluation. The study explained that businesses increasingly rely on automated sentiment systems to capture consumer emotions, assess market reactions, and respond to reputation challenges that happen in real time.

Li et al. (2019) analyses how sentiment in online reviews affects product sales by introducing the Joint Sentiment Topic model (JST) to analyse both the textual content and numerical ratings of customer reviews. They find that positive, sensory topic reviews have a stronger sales impact than reviews focusing on technical attributes. The authors show that both sentiment polarity and aspect-level insights are crucial for predicting sales.

Majumder et al. (2022) investigates the usefulness of customer reviews using sentiment analysis and exploratory data analysis. They analysed reviews on grocery, video games, and digital music using VADER, text mining, and regression modelling. Using the Elaboration Likelihood Model (ELM), they demonstrate that peripheral features such as helpful votes and star ratings significantly impact perceived usefulness. The findings support hybrid models and domain-sensitive tuning, insights that align with the analytical objectives of this thesis.

2.4 Social Media Sentiment and the Automotive Industry

Feature-driven sentiment analysis was proposed by Hu and Liu (2004), who extracted product attributes from customer reviews and linked them to corresponding positive or negative sentiments. This introduced aspect-level sentiment classification, thus allowing businesses to detect the product features that can cause satisfaction and dissatisfaction among customers. This research also proposed a summarization method that transformed thousands of textual reviews into overviews of customer opinions, helping companies to make data-driven improvements.

Chen and Xie (2008) demonstrated that online reviews of customers act as an electronic word of mouth that influences purchasing decisions and market performance. The research showed that the tone and volume of customer feedback significantly affect sales and product reputation. They argued that sentiment-based metrics can enhance traditional marketing indicators by providing real-time insights into customer attitudes, thereby helping to manage the firm's reputation and design effective promotional strategies.

Shukri et al. (2015) conducted sentiment and emotion analysis on Twitter data related to companies like BMW, Mercedes, and Audi. The usage of a Naive Bayes classifier with polarity and emotion lexicons helped them to find that Mercedes is associated with joy while Audi with surprise, by using 3000 tweets, 1000 tweets per brand. Their analysis supported the effectiveness of emotion analysis as an extension of sentiment analysis in competitive industries.

Expanding on regional application of sentiment analysis, Dasgupta et al. (2022) studied sentiment in India's car market during and after the COVID-19 lockdown. They used National Research Council Canada (NRC) emotion lexicons and topic modelling on social platforms like Twitter, YouTube, and news articles. During their research, they found that even though the overall automotive sentiment declined during this period; passenger car sentiment improved due to contactless booking. Their work underscores the importance of sentiment monitoring during crisis-driven market shifts.

In the European automotive context, Zielinski et al. (2023) used a dataset that comprises 1039 German news articles published between February and December 2022, called InfoBarometer, which is a benchmark dataset designed for comprehensible sentiment analysis within the German automotive industry. BERT performed best in classification, but Convolutional Neural Networks (CNN) was superior in interpretability. They focused on sentiments related to socio-technical debates like vehicle electrification.

From a media-focused standpoint, Penmetsa et al. (2023) analyzed how automated vehicles (AVs) are portrayed in the news media with the help of sentiment analysis of over 1.7 million news articles from the time period 2015–2021. The filtration criteria were based on 50 keyword combinations related to AVs and analyzed using VADER, TextBlob, and NRCLex. They discovered negative sentiment on high-profile accidents, and also that political bias influences these sentiments.

Mishra et al. (2024) examined the reputation dimension of 18 automobile brands using VADER and Aspect-Based Sentiment Analysis (ABSA). According to their assessment, Lamborghini and Hummer had the greatest comfort and safety scores. The strength of this paper lies in the wide brand coverage and multi-aspect sentiment visualization, and their ABSA framework highlights how detailed sentiment metrics can inform brand strategy.

2.5 Corporate Reputation and Stakeholder Perception

Fombrun and Shanley (1990) provided one of the earliest empirical examinations of corporate reputation. They claimed that business performance and media visibility both influence the reputation of the firm. They also found that companies engaging in socially responsible behaviour tend to have a better reputation and higher investor confidence. This highlighted that reputation is not static but continuously influenced by public perception and the flow of information.

Extending the understanding of corporate reputation, Barnett et al. (2006) offer a comprehensive analysis on how corporate reputation has been defined by reviewing 49 definitions and classifying them as awareness, assessment, or asset based. This study makes an argument that using the "assessment" view, reputation is seen as collective judgments made by stakeholders about a company's environmental and social performance over time.

The economic and strategic significance of reputation was studied by Gatzert (2015). The study examined how Corporate Social Responsibility (CSR), and stakeholder engagement influence monetary security and market valuation. The results showed that a positive reputation enhances resilience during crises and builds stakeholder trust, while reputation losses can cause negative effects on firm value.

Giachanou et al. (2019) explore how non-sentiment-bearing tweets, which are called polar facts, can affect corporate reputation. They use semantic similarity and supervised lexicon expansion to propagate sentiment and achieve supervised accuracy without labelled data. This study validates the use of sentiments in tracking reputation during crucial events such as Volkswagen's shutdown

Extending this perspective into practical brand monitoring, Vidya et al. (2015) propose a new metric called Net Brand Reputation (NBR) as an alternative to the traditional Net Promoter Score (NPS). With the help of SVM, they classified telecom-related tweets and derived that social media listening provides a dynamic and accurate indicator of brand reputation when compared with traditional surveys like NPS.

2.6 Exploring the Sentiment - Stock Price Relationship

Bollen et al. (2011) analyzed over nine million Twitter posts to examine whether collective public attitude could predict stock market movements. They used opinion lexicons and mood tracking systems like OpinionFinder and Google Profile of Mood States (GPOMS) for emotional dimensions such as happiness, calmness, and alertness. The results revealed that calmer emotions on social media come with positive market movement. This suggests that collective online mood captures investor confidence.

Tirunillai and Tellis (2011) used a four-year dataset covering more than 347,000 product reviews to investigate whether User Generated Content (UGC), which is referred to as online chatter, can be correlated with stock market performance. Through their study, they found that negative sentiment strongly predicts lower returns and higher volatility. This paper thus establishes a foundational link between sentiment and financial indicators, which is considered the core concept of the thesis.

In a different industry, He et al. (2013) applied text mining to selected data from major leading pizza brands on platforms such as Facebook and Twitter. Their finding leads to indicate that employing text mining for strategic decision-making, benchmarking competitors, and real-time social media monitoring are crucial.

Ranco et al. (2015) also studied 1.5 million tweets concerning companies listed in the S&P 500 index. They used event study and correlation analysis to find that Twitter sentiment was closely associated with abnormal stock price returns. Their findings confirmed that real-time social media sentiment can act as an early signal for stock market fluctuations.

In the context of automotive industry, Az-Zahra et al. (2021) study the relation between public emotion and stock price movements for major automotive brands like Volkswagen, General Motors, and Chrysler. The data was obtained through social platforms like Facebook, Twitter, Instagram, and news. The usage of VADER and Pearson correlation analysis helped them to find that sentiment from Twitter and Facebook significantly correlates with stock price, while Instagram does not. This research thus highlights a platform specific strength for stock market forecasting in the automotive sector.

2.7 Ethical considerations in Sentiment Mining

Ethical considerations are considered to be a crucial part when data obtained from social media platforms like Twitter, Reddit, or Facebook is being considered as a vital area of academic discussions and research. Mittelstadt et al. (2016) highlight that there are significant ethical responsibilities when sentiment analytics are used in business and policy decision-making by corporates. Organizations must ensure transparency in how the data are being collected, processed, and interpreted. They must also be concerned about algorithmic systems reinforcing existing social bias. So, when sentiment analysis is used during sensitive corporate events such as factory closures or layoffs, the potential misinterpretations or amplification of misinformation become a crucial part. Contextual accuracy is another concern of sentiment interpretation. Automated lexicon-based tools rely on word polarity and syntactic structure rather than contextual meaning. This can lead to a loss in emotional nuance or in identifying mixed expressions.

Zimmer (2010) studies on Facebook research demonstrate that even anonymized datasets can breach users' expectations of privacy. Researchers should therefore comply with the principles of data minimization and anonymization by only collecting the texts that are necessary for the research purpose and should also avoid redistributing identifiers like usernames that could reveal the identity of the users. In the case of Reddit, the API supports anonymous access to the data that is being retrieved.

These perspectives were complemented by Jobin et al. (2019), by providing a global mapping of AI ethics frameworks and identifying five principles: transparency, responsibility, non-maleficence, justice and fairness, and privacy. Applying these principles to sentiment mining implies that ethical research will have a balance between innovation and individual autonomy.

2.8 Applications in Policy and Stakeholder Decision-Making

The relevance of sentiment analysis expanded from customer research to policy development and stakeholder communication. In the political domain, Stieglitz and Dang-Xuan (2013) demonstrated that the emotional tone in social media content can influence information diffusion and engagement. By analyzing more than 165,000 political tweets, they found that emotionally charged content, both positive and negative, spreads faster and reaches consumers more effectively than neutral content. This implies that sentiment dynamics can shape the agenda-setting process in both political as well as corporate setups. This can assist in developing effective communication strategies during critical corporate events that can lead to reputational decline.

Amangeldi et al., (2024) expanded this to the field of environmental and social policy. They analyzed two years of social media data from Twitter, Reddit, and YouTube to understand public sentiment on climate change and sustainability and found that negative sentiments, which are driven by emotions like fear, distrust, and anticipation, dominate online environmental conversations. These emotional patterns provide early indicators of public anxiety around ecological and industrial transitions.

From the perspective of strategic management, Grande-Ramírez et al. (2022) integrated social media sentiment analysis into a Balanced Scorecard (BSC) framework to support corporate strategic planning. This study showed how key performance indicators (KPIs) linked to external sentiment data enhance decision accuracy and reduce the time required for strategic evaluation. This model demonstrates that integrating public sentiment into organizational dashboards can strengthen the balance between company objectives and stakeholder expectations.

Similarly, Aguilar-Moreno et al. (2024) conducted a bibliometric study revealing that sentiment analysis has evolved into a decision-support mechanism across sectors. This helps in combining aspects like sentiment mining, decision-making, and predictive analysis. The study highlighted how platforms such as Twitter, Reddit, and Amazon provide continuous feedback loops for companies and in policy decision systems.

Finally, Lindgren et al. (2023) described sentiment analysis as a critical method for understanding employee attitudes, internal communication, and policy perception. Their work outlines a three-stage analytical framework consisting of discovery, measurement, and inference. This enabled organizations to translate large-scale text data into insights. They emphasized that sentiment analysis helps identify workforce sentiment during major organizational changes like mergers or downsizing.

2.9 Role of visualization and Analytical Tools in Sentiment Research

Visualization and analytical tools help researchers to interpret large-scale sentiment data in an effective way to gain insights. Keim et al. (2008) describe visual analytics as the integration of automated data analysis with an interactive visual interface that supports human reasoning and insight generation. According to their framework, in order for analytical visualization to be effective, computing capacity and cognitive interpretation must be combined. This enables analysts to investigate and confirm ideas through visual feedback loops. This principle is relevant in sentiment research, as multidimensional data like text polarity and frequency should be transformed into valuable insights.

Ziegler et al. (2010) discuss how visual analytics support the identification of trends and patterns in financial time series data. Their approach includes interactive dashboards that combine statistical indicators with visualization to enhance situational awareness and decision-making. Lindgren et al. (2023) applied sentiment analysis within organisational research and presented a three-stage analytical model. Insights on employees' emotions and communication tone were obtained from large-scale textual data using visualization. This demonstrated the importance of visualization within the social analytical workflow to understand and gain insights on sentiments during organizational transitions. In the work of Grande-Ramírez et al. (2022), visualisation of external sentiment indicators improves the accuracy and timeliness of management decision making.

Bell et al. (2024) focus on the role of sentiment visualisation where social media data was employed to demonstrate how sentiment dashboards show shifts in consumer tone before a crucial reputational event happens. Collectively, these studies reveal that visual analytics dashboards can act as a bridge between computational models and managerial decisions.

2.10 Sentiment Analysis for Reputation and Brand Monitoring

Vidya et al. (2015) proposed the NBR model, where sentiment data was integrated with NPS. This approach showed that sentiment-based monitoring provides an accurate reflection of brand health when compared to traditional survey results. Through their studies, they found that NBR fluctuated responsively to customer attitudes more than the conventional way of finding the reputational indices, thus highlighting the importance of social media data monitoring.

Similarly, Bell et al. (2024) demonstrated the application of social media sentiment for real-time reputational management. According to their analysis, negative sentiment increases regularly followed falls in share-of-voice and engagement, which acted as early warning indicators for public relations teams. This supports that the use of sentiment dashboards helps in detecting and mitigating reputational crisis before they escalate.

Kaul et al. (2016) from a strategic point of view, integrated the case analysis of 12 global corporates and examined how social platforms and analytical tools were integrated into their corporate structure to monitor stakeholder narratives. By cross-comparison and analysis, they showed that companies that incorporate sentiment feedback into their communication strategies have a faster recovery from public relations crises and are able to achieve a higher level of stakeholder trust. In addition, Aguilar-Moreno et al. (2024) conducted a bibliometric review of sentiment analysis on business decision-making. They found a clear shift towards using sentiment indicators for reputational management. These indicators were used as components of strategic systems that integrate reputation metrics with corporate administration and stakeholder analysis.

2.11 Public Sentiments and Financial Markets

Early empirical studies by Bollen et al. (2011) established the relationship between public sentiments and financial markets by analysing over nine million Tweets to examine whether the collective mood states could predict fluctuations in the Dow Jones Industrial Average (DJIA). This research provided large-scale evidence that aggregated online emotions can influence investor psychology and market behaviour.

Later on, Tirunillai and Tellis (2012) analysed a dataset of four years and investigated how online-generated content can affect stock performance. Their studies found that online chatter acts as an early indicator of investor sentiment and thus adds to the financial models in that organisation. Ranco et al. (2015) through event study and correlation analysis, found abnormal surges in Twitter sentiment and attributed these to short-term excess stock returns. This confirmed that social media activity can act as a real time sensor for market information.

Similarly, Chan et al. (2017) examined firm stock data and concluded that pricing dynamics are strengthened by behavioural signals influenced by public perception and media tone. Az-Zahra et al. (2021)used correlation analysis of social media sentiment and stock prices for companies to discover that sentiment extracted from some media showed significantly positive correlation with short-term stock movements compared to other selected media. This shows the importance of choosing the appropriate social media platform for financial forecasting and reputational studies. All these studies illustrate that online sentiment serves both diagnostic and predictive functions in financial analysis.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Research Approach

A quantitative, exploratory research design was employed to examine the potential relationship between customer sentiments and the company's stock price, which served as a proxy for reputation. The methodology integrates NLP for sentiment analysis (Nasukawa and Yi, 2003) and financial time series analysis to assess the reputational implications of the company's corporate decision (Ziegler et al., 2010).

3.2 Data Sources

1. Reddit based Sentiment Data

Sentiment data collected from Reddit posts and user comments related to Volkswagen factory shutdown and layoffs. Reddit was chosen for its long form discussions, focused subreddits which can offer a better dataset for analysing real time public opinion (Reddit., 2025).

2. Stock Market Data

Stock price data of Volkswagen AG (VOW3.DE) retrieved from Yahoo Finance, serving as the financial proxy for company reputation (Yahoo Finance., 2025).

3.3 Data Collection

3.3.1 Reddit Data Collection

(a) Reddit as a social media platform for Sentiment and Reputational Analysis

Reddit is a digital community platform which has a system of topic-based groups known as "subreddits." Here, users discuss topics, share content, and react to current events. This was initially a content-sharing site which then evolved into a self-referential discussion network (Singer et al., 2014). Reddit reports more than 500 million active users daily as of the second quarter of 2025, with a significant percentage of younger demographics participating (Backlinko, 2025; Exploding Topics, 2025). Due to its decentralized structure, topic-driven interaction, rich textual data, real-time conversation, and high degree of anonymity, the platform is perfect for conducting sentiment analysis research on brand reputation and for gathering public opinion during corporate events. Additionally, Reddit has introduced advanced modern tools and AI-enhanced features that ensure discussion quality and relevance (Reddit Inc., 2025). Thus, this platform can serve at both quantitative as well as qualitative levels to explore more on emotional and opinion-based reactions to events like Volkswagen's factory shutdown.

(b) Subreddits and Keywords Selection

Subreddits on Reddit are topic-specific communities where the selection of the right one is essential for relevance to the Volkswagen factory shutdown. For this study, the data is collected between a time frame of October 1, 2024, and February 15, 2025. However, due to a high frequency of missing data and a decline in comment engagement after December 2024, the final dataset was considered between October 1, 2024, till December 31, 2024. This period was chosen strategically to record the public opinion and online conversations that can happen pre and post the official closure announcement on December 12, 2024. This helped in evaluating the user response throughout the event that was being considered. The chosen subreddits cover a wide range of areas including automotive industry discussions, regional news, economic implications, and so on. A total of 10 subreddits were chosen for the data scraping process, which includes: Volkswagen, cars, autos, news, Europe, Germany, electricvehicle, worldnews, autonews, and AskEurope. These subreddits were selected based on the probability of including user-generated content that discussed Volkswagen, the automotive industry, labour issues, and so on within Europe and Germany.

Simultaneously, a detailed list of 20 keywords and phrases was defined to scrape the posts within these subreddits. These keywords reflected various aspects of the shutdown event, including the employment implications that might occur if this decision comes into play, production halts, plant closure, and industry shutdown. Some of the keywords that were used are: “Volkswagen factory shutdown,” “VW layoffs,” “Audi Brussels shutdown,” “Volkswagen to close factory,” and “auto industry layoffs.” These keywords were selected by reviewing the news headlines, Reddit post titles, and also by experimenting with the topic patterns that can happen to cover the ways in which the discussion about this incident may unfold. The obtained data is then used to perform sentiment analysis to understand more about the public opinion on the occurred event.

3.3.2 Stock Price Data Collection

The stock data of the selected company (Volkswagen) was obtained from Yahoo Finance using the yfinance Python library (yfinance, 2025), which is widely used for the extraction of financial data in research. Yahoo Finance was chosen as the primary source of data because of its detailed, publicly accessible, and free financial market information. This platform also integrates well with Python libraries, which makes it easy to scrape and analyse financial data (Yahoo Finance, 2025).

This study uses the ticker symbol VOW3.DE, representing Volkswagen AG on the Frankfurt Exchange. The stock data was extracted for the period from July 1, 2024, to April 15, 2025, which helps in covering the fluctuations that might occur before and after the factory shutdown announcement in December 2024. Thereafter, the data is resampled to align with the sentiment scores that are generated from the data collected from Reddit, to understand the public sentiment and the financial aspect, which in turn depicts the reputation of the company.

	A	B	C	D	E	F
1	Date	Close	High	Low	Open	Volume
2		VOW3.DE	VOW3.DE	VOW3.DE	VOW3.DE	VOW3.DE
3	2024-07-01 00:00:00	106.05	107.6	106.05	106.8	783736
4	2024-07-02 00:00:00	105.75	106.1	104.4	105.85	948029
5	2024-07-03 00:00:00	106.5	107.15	105.8	106	957213
6	2024-07-04 00:00:00	107.6	108.25	106.85	107	522707
7	2024-07-05 00:00:00	106.85	108.45	106.55	107.7	711334
8	2024-07-08 00:00:00	106.8	107.3	106.2	107	483947
9	2024-07-09 00:00:00	106.6	107.25	105.95	106.25	619664
10	2024-07-10 00:00:00	106.25	108.5	104.2	106.9	1220032
11	2024-07-11 00:00:00	107.15	107.8	106.2	106.75	869082
12	2024-07-12 00:00:00	108.55	108.9	107.4	107.55	772086
13	2024-07-15 00:00:00	108.2	108.4	107.65	108.3	499306
14	2024-07-16 00:00:00	106.95	107.8	106.35	107.6	808621
15	2024-07-17 00:00:00	106.45	106.85	105.5	106.55	520279
16	2024-07-18 00:00:00	107.6	108.45	107	107.2	791215
17	2024-07-19 00:00:00	105.6	106.8	105.25	106.55	985836
18	2024-07-22 00:00:00	106.65	107.1	105.65	105.7	633234
19	2024-07-23 00:00:00	105.65	106.45	104.25	105.65	931600
20	2024-07-24 00:00:00	105.35	105.8	104.35	104.4	548776
21	2024-07-25 00:00:00	104.65	104.65	102.15	103.5	1302332
22	2024-07-26 00:00:00	104.8	106.1	102.9	103.2	901865
23	2024-07-29 00:00:00	104.3	105.85	104.25	105.35	788471
24	2024-07-30 00:00:00	104.05	105.05	103.75	104.5	696927
25	2024-07-31 00:00:00	103.2	104.35	102.25	103.5	1191151
26	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Figure 3.1: Excel Screenshot of Volkswagen stock price dataset extracted from Yahoo Finance

Figure 3.1 represent the Excel sheet with the daily trading information of VW AG extracted using yfinance library. This captures the company's market activity on the Frankfurt Stock Exchange.

It contains six key variables which are:

- Date: the specific trading day record
- Open: the stock price at the beginning of the trading day
- High: the maximum price reached during that day
- Low: the minimum price recorded for that day
- Close: the final closing trade price for that day which is used as the performance indicator
- Volume: the total number of shares traded on that day.

3.4 Data Preprocessing

3.4.1 Reddit Data Preprocessing

The retrieved initial dataset from Reddit contains user comments related to Volkswagen's factory shutdown event. The dataset is then subjected to NLP techniques, which are used to clean, normalize, and prepare the text for sentiment analysis. The processes applied to the dataset are:

(a) **Text Normalization:** the comments are converted to lowercase to maintain uniformity, which helps in reducing linguistic variability due case sensitivity.

(b) **URL Removal:** to remove noise, hyperlinks embedded in the comments are removed, as they didn't contribute to any semantic value to the analysis.
(c) **Punctation and Symbol Removal:** to retain meaningful content, punctations, special symbols ad non alphanumeric characters are removed.
(d) **Duplicate Removal:** duplicate comments were removed to eliminate sampling bias, the dataset was then reduced from 18,625 raw comments to 1,533 unique entries after deduplication.
(e) **Stop Word Filtering:** common English words like "and", "the" and "is" were removed using the Natural Language Toolkit (NLTK) stopword list, ensuring that only texts that are relevant to the context are retained.
(f) **Language Filtering:** only English language comments were considered to ensure consistent sentiment interpretation.

The cleaned text is then passed through the VADER sentiment analyser, where each comment receives a compound score between -1 and +1. The comments are then categorised as positive (≥ 0.05), neutral (between -0.05 and 0.05), and negative (≤ -0.05), following the VADER interpretation threshold (Hutto and Gilbert, 2014).

	A	B	C	D	E	F	G	H	I	J
1	post_id	comment	score	created	neg	neu	pos	compound	sentiment	title
2	1fc23oa	all is well with this bu	43	2024-09-08 18:52:21	0.275	0.619	0.107	-0.5334	negative	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
3	1fc23oa	the media has been p	22	2024-09-08 19:18:33	0.067	0.893	0.04	-0.9154	negative	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
4	1fc23oa	i read a lot of reactio	6	2024-09-08 22:51:16	0.154	0.661	0.185	0.2382	positive	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
5	1fc23oa	media rides high on h	2	2024-09-08 19:32:47	0	1	0	0	neutral	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
6	1fc23oa	this is the company t	5	2024-09-08 19:31:18	0.141	0.742	0.117	-0.5346	negative	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
7	1fc23oa	vw needs to bring the	3	2024-09-08 23:21:06	0	1	0	0	neutral	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
8	1fc23oa	the results of declinir	10	2024-09-08 18:52:15	0.054	0.902	0.044	-0.1027	negative	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
9	1fc23oa	spambot	2	2024-09-08 19:34:57	0	1	0	0	neutral	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
10	1fc23oa	another aspect of thi	2	2024-09-08 19:45:51	0.094	0.821	0.086	-0.4003	negative	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
11	1fc23oa	vw needs to come dc	6	2024-09-08 19:27:26	0.135	0.839	0.026	-0.9194	negative	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
12	1fc23oa	i wonder how many b	1	2024-09-08 23:58:27	0.096	0.827	0.077	-0.1838	negative	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
13	1fc23oa	$77,000 for an id buz	1	2024-09-09 01:29:54	0.194	0.688	0.119	-0.296	negative	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
14	1fc23oa	i don't quite understa	1	2024-09-09 15:04:08	0.072	0.841	0.087	0.128	positive	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
15	1fc23oa	and people downvote	14	2024-09-08 19:26:55	0	1	0	0	neutral	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
16	1fc23oa	i've visited car factor	2	2024-09-09 00:22:17	0.059	0.842	0.1	0.7003	positive	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
17	1fc23oa	also every article say	1	2024-09-09 02:07:25	0	1	0	0	neutral	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
18	1fc23oa	all honda and vw nee	1	2024-09-09 09:34:59	0.14	0.86	0	-0.3818	negative	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
19	1fc23oa	yeah. the company th	1	2024-09-08 23:18:56	0.031	0.896	0.073	0.2846	positive	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
20	1fc23oa	agree they need the g	1	2024-09-09 00:26:33	0.026	0.87	0.105	0.8001	positive	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
21	1fc23oa	couldn't agree more	3	2024-09-08 18:58:38	0.072	0.839	0.089	-0.0772	negative	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
22	1fc23oa	like the shitty taos th	2	2024-09-08 23:21:15	0.304	0.591	0.105	-0.6908	negative	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
23	1fc23oa	i would argue that vw	0	2024-09-08 19:57:00	0.245	0.755	0	-0.7964	negative	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
24	1fc23oa	they also learnt how	1	2024-09-08 21:04:12	0	1	0	0	neutral	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM
25	1fc23oa	that's an absolutely i	2	2024-09-09 03:33:25	0.272	0.728	0	-0.4576	negative	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES AND JOB CUTS LOOM

Figure 3.2: Excel Screenshot from the cleaned Reddit comment dataset after VADER sentiment analysis.

Figure 3.2 illustrates a portion of the processes Reddit comments collected from subreddits related to the company event. Each record includes unique post ID, comment text, engagement score, timestamp, sentiment metrics which are generated using VADER model.

3.4.2 Stock Market Data Preprocessing

The preprocessing steps that are done to the initial stock market dataset of Volkswagen, obtained from Yahoo Finance, are:

(a) Range Filtering: The data was filtered to a period from 1st July 2024 till 15th April 2025, to cover the pre, during, and post factory shutdown announcement.
(b) Data Selection: The analysis focused on the closing price for each trading day, which is the stock's final price at market close.
(c) Resampling to Weekly Frequency: The daily stock closing process wee resampled into weekly averages. This allowed for a consistent comparison with aggregated weekly sentiment scores when that dataset is obtained.

3.5 Sentiment Analysis

Understanding the sentiments of the public surrounding Volkswagen's 2024 factory closure was one of the main objectives, where the sentiment analysis was executed using the VADER model. VADER is a lexicon- and rule-based framework designed for effective analysis of brief, informal text, including social media content (Hutto and Gilbert, 2015). This tool is effective for capturing emoticons, capitalization, and punctuation-based emphasis, which is suitable for a social media platform like Reddit.

Each extracted Reddit comment was analyzed using the Sentiment Intensity Analyzer from the Natural Language Toolkit (NLTK). The tool generated four sentiment scores: positive, negative, neutral, and compound. The compound score, ranging from -1 (most negative) to +1 (most positive), was used to classify comments into three categories.

(a) Positive, when the compound score is greater than 0.05

(b) Negative, when the compound score is less than -0.05

(c) Neutral, when the score falls within the range -0.05 to 0.05

These scores were incorporated into the cleaned Reddit dataset, and the weekly averages were computed to identify the trends over time. While domain-specific sentiment tools can provide better precision, VADER demonstrates strong performance on social media text, making it suitable for this application without requiring additional lexicon training (Gräbner et al., 2012).

3.6 Merging Data and Correlation Analysis

The cleaned sentiment and stock price datasets were combined using the pandas.merge() function in Python, based on matching weekly time indices in order to examine the connection between customer sentiment and financial performance. Resampling both datasets to weekly frequency ensures that each row represents a specific week with two aligned indicators from both datasets. Each week, a single row in the generated output includes the company's weekly average stock closing price as well as the weekly average compound sentiment score.

Date	Weekly Sentiment	Weekly Stock Price
2024-07-14 00:00:00	0.015509942	107.0700012
2024-07-21 00:00:00	0.5106	106.9599976
2024-07-28 00:00:00		105.4200012
2024-08-04 00:00:00		101.5180008
2024-08-11 00:00:00		93.8960022
2024-08-18 00:00:00		93.80000153
2024-08-25 00:00:00		96.47200012
2024-09-01 00:00:00		96.44400024
2024-09-08 00:00:00	0.064225632	95.08799896
2024-09-15 00:00:00	0.08356875	90.39199982
2024-09-22 00:00:00	0.254975	92.13600006
2024-09-29 00:00:00		94.45600128
2024-10-06 00:00:00		93.33200073
2024-10-13 00:00:00		93.10000153
2024-10-20 00:00:00		91.55200043
2024-10-27 00:00:00		91.7
2024-11-03 00:00:00	0.029088201	89.61999969
2024-11-10 00:00:00	0.136875	86.11600037
2024-11-17 00:00:00	0.4404	83.52400055
2024-11-24 00:00:00	-0.05045	82.34400177
2024-12-01 00:00:00		80.72000122
2024-12-08 00:00:00	0.61135	81.54400024
2024-12-15 00:00:00	-0.6476	85.92799683
2024-12-22 00:00:00	0.040665625	87.33200073

Figure 3.3: Excel Screenshot of Merged Weekly Sentiment and Stock Dataset

Figure 3.3 shows the glimpse of merged dataset that integrated weekly average sentiment scores with VW's weekly average stock prices. This integrated dataset forms the input for the correlation analysis and Power Business Intelligence (Power BI) visualization.

Furthermore, with the Pearson correlation analysis, the strength and linear relation between the weekly average stock prices (dependent variable) and the weekly average sentiment scores (independent variable) were assessed. The correlation coefficient ranges from -1 to +1:

(a) r greater than 0 indicates a positive linear relationship
(b) r less than 0 indicates a negative relationship
(c) r equals to 0 suggest no linear correlation

The statistical significance of the relationship between sentiment and stock performance was then ascertained by computing the p-value, where the threshold of p was used to determine whether the relationship was statistically meaningful. Thus, correlation can help in assessing whether the shift in public sentiment around an event like a factory shutdown, in the case of a company like Volkswagen, was reflected in stock price fluctuations.

3.7 Timeline Segmentation

For better interpretation, the merged dataset was also segmented into four event phases based on timeline of Volkswagen's factory shutdown event. This segmentation allows a clear understanding of how sentiment patterns and stock trends evolved during each critical stage of the corporate event. This analysis provides insights into whether sentiment fluctuations align with shifts in the company's strategic decisions (Tirunillai and Tellis, 2011). These segmented timelines are later on mentioned as the event markers.

The event phases were defined as follows:

(a) Speculation Phase (September 1 to October 28, 2024): where early media speculations and rumours began circulation about the plant closures which was prior to official announcement (Waldersee and Amann.,2024).

(b) Negotiation Phase (October 28 to December 20, 2024): active discussions between company, labour unions and government bodies regarding the future of the factory happened in this phase (Ziady., 2024).

(c) Post Agreement Phase (December 20, 2024, onwards): refers to the Volkswagen's official statement on factory shutdown (Euronews., 2024).

(d) Others, referring period outside these ranges.

3.8 Data Visualization

To ensure clear understanding and to gain more insights from the analysis, a range of visualizations was developed using Microsoft Power BI. These visualizations are used to depict various analytical dimensions of the dataset.

1. **Line charts** were generated to depict the progression of the weekly average sentiment score. These charts incorporated event phase markers, helping in the mapping of sentiment fluctuations.
2. **Pie charts** were used to display the overall distribution of sentiment labels (positive, negative, and neutral) assigned to the cleaned Reddit comments. This allowed a comparative view of the proportion of public opinion across sentiment categories.
3. **Combination charts** (dual-axis line charts) were used to simultaneously plot weekly sentiment scores and corresponding weekly average stock prices. This helps to visualize the alignment between public response and market performance during key event phases.
4. **Scatter plots** were produced to visualize the correlation between sentiment and stock performance. These serve as a depiction of the statistical analysis used to assess the significance of the relationship.

3.9 Tools and Technologies

The analytical workflow of this study was implemented using a combination of programming tools, data libraries, and visualization platforms.

1. Python 3.11.7 and Jupyter Notebook

Python (Python Software Foundation, 2024) was used as the primary programming language due to its versatility and the extensive libraries available for data analytics applications**.** Jupyter Notebook (v 7.0.8**)** (Project Jupyter Team, 2025) was used for analysis which allowed inline visualization and stepwise code execution. This environment supported iterative testing and debugging throughout the workflow.

Key functions included:

- Executing data pipelines for retrieval of Reddit comments using **Python Reddit API Wrapper (PRAW)**.
- Running **VADER** sentiment scoring within loops that parsed each comment.
- Generating intermediate exploratory charts such as sentiment histograms and scatter plots using **matplotlib.pyplot** for verification purposes.

2. PRAW (v7.7.1)

Relevant conversation threads from specific subreddits, including **r/Volkswagen**, **r/cars**, **r/auto**, and **r/europe**, were gathered using PRAW. The **search()** method of this module allowed for subreddit targeting, metadata gathering, and keyword-based filtering. The **comments.list()** method was used to retrieve related comments for each article, and a Pandas dataframe was used to record metadata including comment score, timestamp, and anonymized authors ID. Real-time and context-specific data collection was guaranteed by this API-based procedure, which precisely reflected the online conversations around the business event (Boe, 2023).

3. Pandas (v2.1.4)

This Pandas handled all tabular operations and in data manipulation tasks such as comment deduplication, timestamp parsing, weekly resampling, data merging and transformation of sentiment scores. (pandas development team, 2025)

Function used included

- **drop_duplicate(subset =’comment_body’)** was used to remove the redundant text entries
- **pd.to.datatime()** converted Unix timestamps to Readable formats
- **resample(‘W’)** generated the weekly averages of the sentiment scored with the Volkswagen stock timeline
- **pd.merge()** was used to join the sentiment and the stock data for correlation analysis
- **groupby(‘sentiment_label’)** grouped comments by polarity labels that helped in getting the frequency counts.

4. NLTK (v3.8.1) and VADER

The VADER sentiment model, integrated with the NLTK framework, was used to classify comments into positive, negative, and neutral categories using compound sentiment scores generated by the **SentimentIntensityAnalyzer()** function. This generated polarity matrices including **pos**, **neu**, **neg**, **compound**.

Each compound score was labelled based on the following predefined thresholds:

- Greater than or equal to 0.05 is given as positive
- Less than or equal to -0.05 is given as negative
- Between -0.05 and 0.05 is given as neutral

5. Microsoft Excel

Excel was used as an intermediate tool for storing, viewing, and exporting the processed dataset. The cleaned data from Python was stored in Excel using .xlsx format and then imported to Power BI for visualization (Microsoft Corporation, 2025).

6. Microsoft Power BI

Used to create dynamic and interactive data visualizations, including line charts for sentiment analysis, pie charts for sentiment distribution, combination charts for sentiment-stock alignment, and scatter plots for correlation interpretation (Microsoft Corporation, 2025).

CHAPTER 4: RESULTS AND ANALYSIS

This chapter presents the analysis of the sentiment scores obtained from the Reddit dataset after applying VADER sentiment analysis. A correlation study was also conducted between the sentiment data and Volkswagen's historical stock price data. The key objective of this analysis was to gain insights on how the factory shutdown announcement made in December 2024 impacted public sentiment, and how these sentiment patterns may have influenced the company's reputation, represented through its stock performance. A series of visualizations were developed using Microsoft Power BI to illustrate temporal sentiment trends, sentiment distribution, sentiment–stock correlation, and relevant qualitative insights. The selected analysis timeline spans from July 1, 2024, to December 31, 2024, and is segmented into four event phases based on key milestone developments.

4.1 Sentiment Distribution and Top Commented Post Analysis

(a) Sentiment Distribution

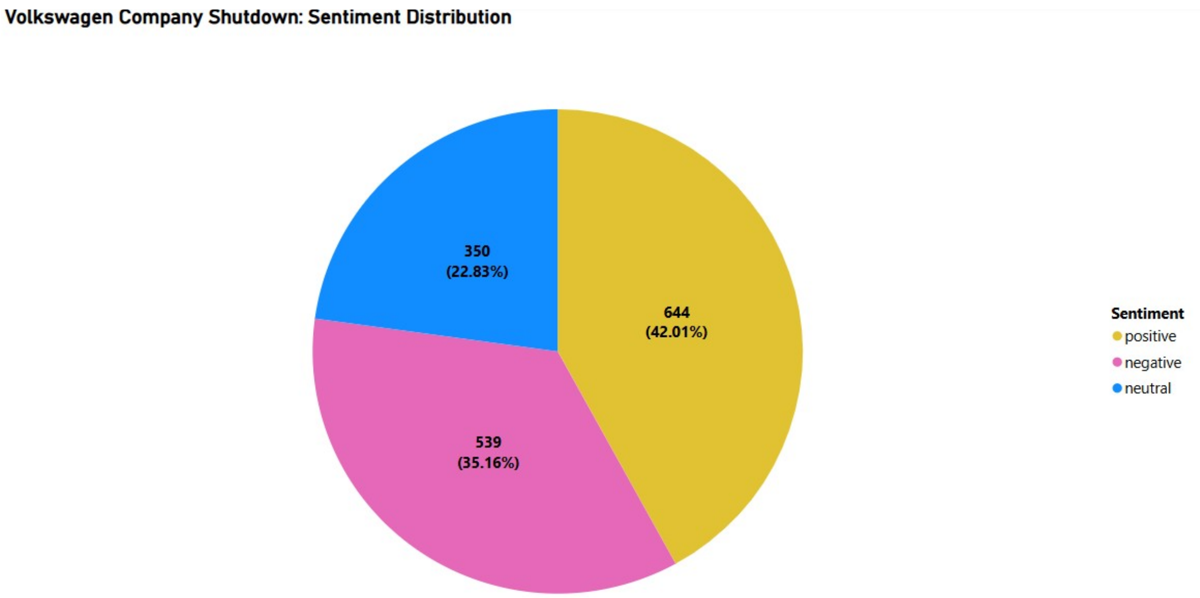

Figure 4.1 Sentiment Composition on Reddit – Volkswagen Shutdown Event

Figure 4.1 illustrates a pie chart with the distribution of the sentiment across all Reddit comments that are related to the announcement of Volkswagen's factory shutdown in 2024. Using VADER sentiment analysis, the comments were divided into three groups: positive, negative and neutral. Out of 1,533 distinct comments collected, 42.01% (644 comments) were positive, 35.16% (539 comments) were negative, and 22.83% (350 comments) were neutral.

This distribution suggests that, despite the expectations around the event of factory shutdown, the overall public reaction on Reddit was not predominantly negative. As shown in Figure 4.1, positive comments cover a large portion, suggesting that the public responded to the shutdown decision with support, trying to understand the long-term strategic objectives of the firm. Therefore, this could be used as an indicator of optimism around long-term strategic alignment of the management.

As per the analysis, one-third of the sentiment with negative comments can include issues like job loss, economic impact, and loss of transparency in communication. The proportion of the positive and negative content can lead to the assumption that the customers were not overwhelmed by the overall event happening. About 23% of the comments were neutral, showing that there can be customers who were not expressive about an emotional opinion. This can include comments that report the news, shared articles, or comments with questions without including an opinion about the event.

The relatively balanced sentiments depict the balanced nature of public sentiments which did not provoke a reputational crisis at the time of Reddit discussions. Rather, it created a space for a range of responses by offering neutral observation, criticism, and support. Instead of a one-sided negative backlash, which is usually expected when a huge corporation like VW announces downsizing, these overall insights show how balanced public perception is.

(b) Top Commented Post Analysis

```
🔝 Top Posts by Sentiment Category:
                                              title sentiment  Comment_Count
24  Volkswagen says considering factory closures i...  positive            220
22  Volkswagen says considering factory closures i...  negative            144
23  Volkswagen says considering factory closures i...   neutral            108
```

Figure 4.2: Top Reddit Post by Sentiment Classification

Figure 4.2 shows the top posts by sentiment category of positive, neutral, and negative comments. Notably, the post titled “Volkswagen says considering factory closures in Germany” served as the central point of public engagement. This post obtained 220 positive, 144 negative, and 108 neutral comments, which reflects the multidimensional emotional response on the same post. This shows how the users interpret corporate announcements, as some have perceived it as a restructuring effort from the company which can have a positive impact, while some signal it as a loss which depicts the negative expression through comments, while some consider this as a factual update which contributed to the neutral expressions in the comments.

The presence of a single post as the top comment for all three sentiment categories shows how strongly this post resonated with the public and became a focal point of discussion among other posts. This illustrates how a single high-impact post can include the emotional complexity of a stakeholder, while at the same time also showing the different emotional aspects in which the post is perceived by the customers depending on the personal and economic context.

From an analytical perspective, this shows the importance of tracking the patterns in trending posts during a crucial event can help the company to gain real-time insights into the public opinion as well as into the emotional aspect of that opinion.

4.2 Weekly Sentiment Trend by Event Phases

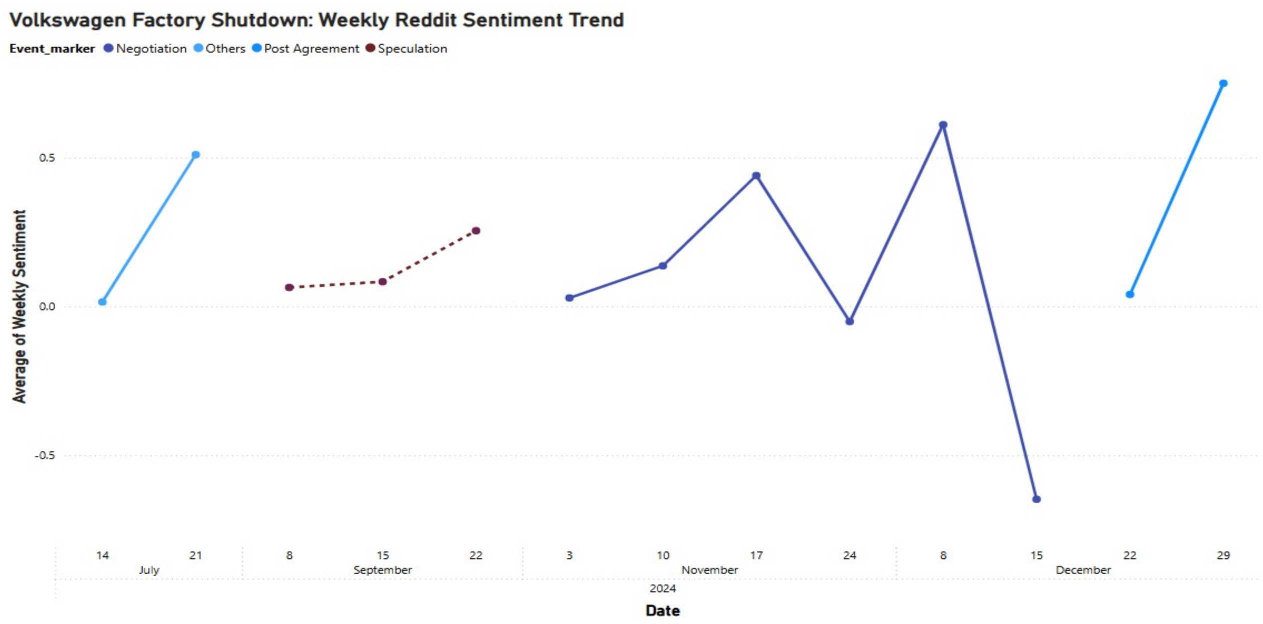

Figure 4.3: Weekly Reddit Sentiment Trend with Event Based Variation

Figure 4.3 illustrates a line chart for the average weekly sentiment of collected Reddit comments discussing the company shutdown. The data was collected from a period of July till December 2024. The main event that is focused on in this study was speculated in early September. The above-mentioned time period was divided into four phases, and it is used in this chart for better insights.

(a) Others (From July till August 2024)

This phase serves as a baseline to understand the public opinion before the event take place thus providing a reference point for comparison. Sentiment starts off as neutral (~ 0.0) and rises to approximately +0.5, suggesting that at the early stage there was curiosity, optimism, and low public concern unrelated to the shutdown fears. The rise in positive emotion shows the brand image that it has in its respective market. This shows that there were healthy discussions happening about the company on this social media platform which weren't affected by the upcoming event. This stability and positivity during this phase can reinforce that future changes in sentiment are likely event driven rather than a part pf the ongoing trend.

(b) Speculation Phase (September till Late October 2024)

This period highlights the power of media speculation even before the official confirmation of the factory shutdown happens. The fluctuation in sentiments during this phase shows the sensitivity of stakeholders to rumours or on any leaked information even without official confirmation. During this phase, there is a significant dip in emotion when compared to August. The sentiment drops to ~ 0.2, reflecting the first sign of concern following the speculation in media, thus spreading to social media platforms like Reddit. This suggests that even before the official announcement of the factory shutdown and layoffs, the speculative discussions were enough to create concerns and reduce public optimism. It then rises slightly to ~ 0.3, indicating a mild emotional shift. Even though the sentiment did not become negative, the sudden fall from the positive chatter depicts a reputational risk as this speculation gained public attention. Compared with the earlier neutral to positive tone of the public, this drop in this phase depicts a shift in the public's emotional state reflecting the early reputational impacts.

(c) Negotiation Phase (November till Mid December 2024)

The contrast between peak optimism and sharp negativity show the unpredictability in the emotional aspects of the customers concerned with the event happening. This can be due to inconsistent communication or unmet public expectations on the negotiations initiated by the firm. From Figure 4.3, the emotional climb observed in this phase, where the sentiment increases above +0.5, which can be a possible response to the negotiation developments, management response, or public dialogue suggesting compromises. But later on, there is a sharp drop below -0.5, showing discontent in the confirmed closure plans and the cost-cutting methods taken by the company. This drop marks the most negative point, highlighting the reputational vulnerability during the decision-making period for Volkswagen. This evident fluctuation suggests that while transparent negotiations can offer reassurance, sudden confirmation of unfavourable outcomes can undo the positive sentiment gains.

(d) Post Agreement (Late December till Early January 2025)

There is a significant increase to ~ +0.58, which is likely due to the final agreement terms being adapted by the company regarding the closure. This may indicate relief among stakeholders, which is found common in crisis recovery narratives where uncertainty is resolved. It might also represent a shift in focus from criticism to adaptation as well.

4.3 Sentiment versus Stock Price Over Time

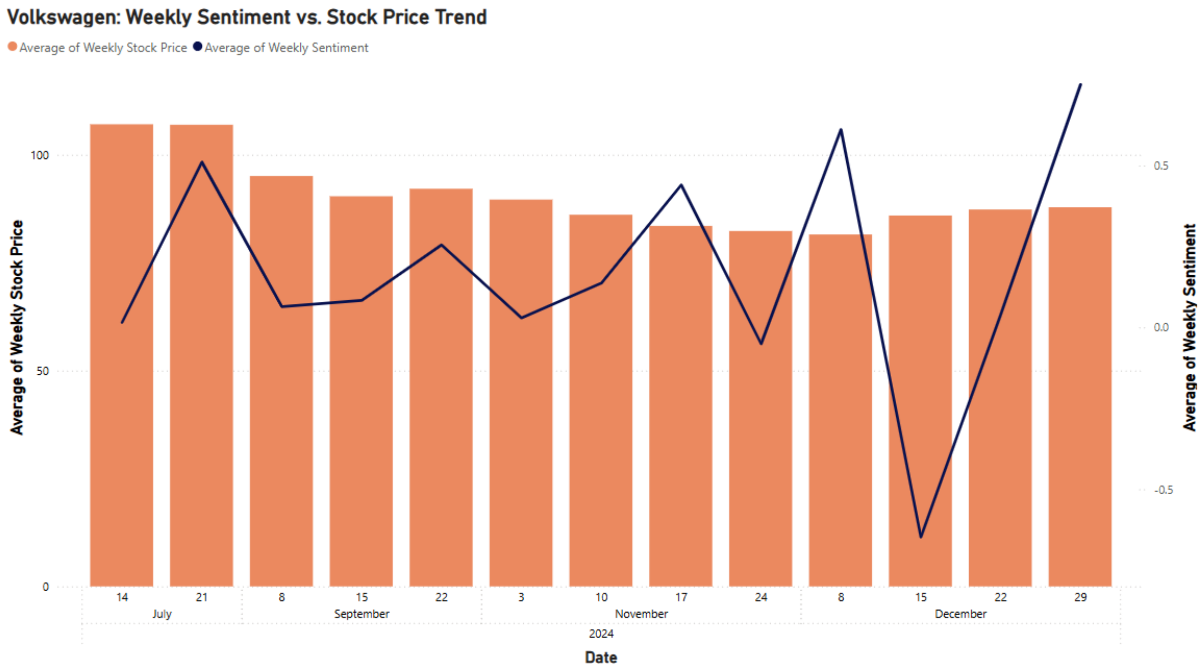

Figure 4.4 Weekly Sentiment and Stock Price Trend

Figure 4.4 is a combo chart that visualises weekly average sentiment score of the retrieved Reddit comments as a line graph (right Y axis) and the company's stock price in a bar graph (left Y axis) over the period July to December 2024. This visualization helps in understanding both emotional and financial reactions to the company's shutdown announcement over time.

In the early period (July – August), the stock price remains relatively high, around €108-€110, where the sentiment rises significantly with a peak near +0.48, indicating a stable market with a positive or neutral public discussion. This stage serves as a baseline of the brand perception by recording the natural conversations unrelated to the shutdown event.

During the speculation phase (early September to late October), the stock price begins to decline gradually, dropping to around €95-€97, and the sentiment also seems to drop steeply from +0.48 to around +0.02. This can indicate emotional disturbance following the speculation of the factory shutdown and also the uncertainty in the investor's behaviour.

In the negotiation phase (November to mid-December), sentiment fluctuates sharply. Initially, the average sentiment rises above +0.5, possibly reflecting public trust in the ongoing negotiations. However, as the official confirmations regarding the company's closure are announced, the sentiment suddenly drops to below -0.5, indicating strong dissatisfaction with the decisions. Meanwhile, the stock price continues to decline and then becomes relatively stable between €88–€95, highlighting a disconnect between public sentiment and market response.

As the post-agreement phase approaches (late December), the sentiment recovers slowly with a peak of approximately +0.6, which can reflect optimism about the new measures taken by the company, and the stock price remains largely unchanged around €91, indicating limited immediate emotional reaction from the market.

Overall, the insights suggests that although sentiment is sensitive to corporate announcements, it doesn't show a direct synchronous relationship with the stock market performance, which is chosen to be a proxy for the company reputation.

4.4 Correlation Analysis

This section focuses on understanding more about the statistical relationship between the weekly average sentiment score, which was retrieved from Reddit, and the company's stock price performance. Pearson correlation analysis was applied to two timeframes: one covering the entire analysis period from July to December 2024, and the other specifically focusing on the Speculation Phase, which was from September 1 to October 28, 2024.

The hypothesis that was considered

H_0: There is no significant relationship between customer sentiment on social media and Volkswagen's stock price.

H_1: There is a significant relationship between customer sentiment on social media and Volkswagen's stock price.

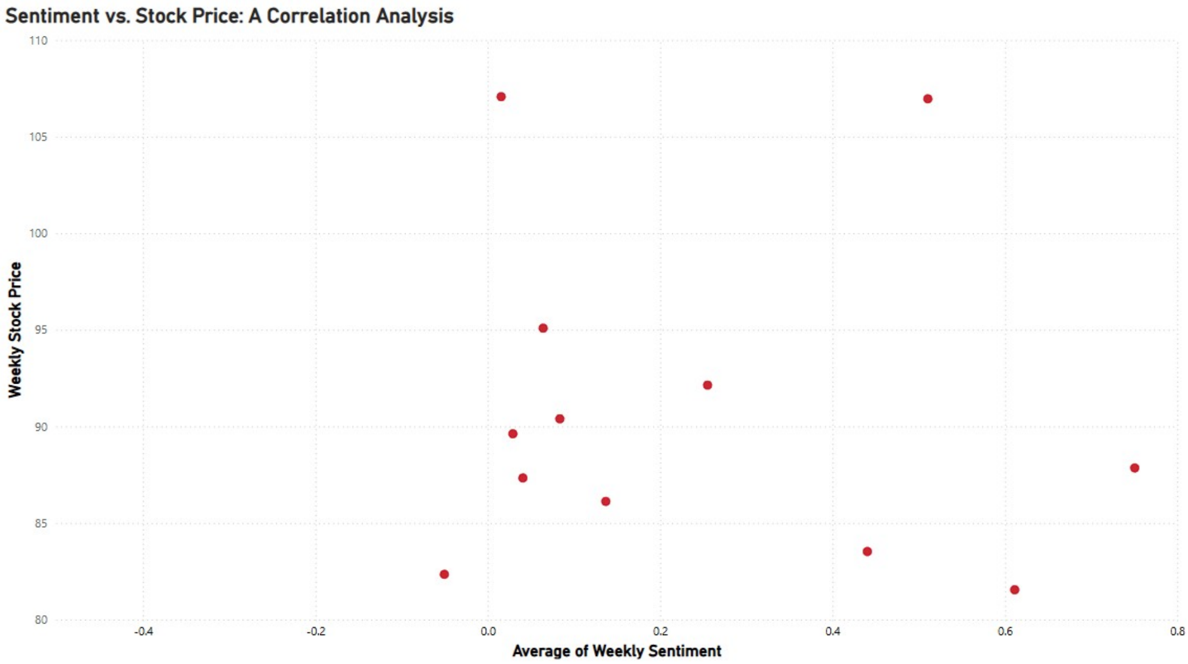

Figure 4.5: Scatter Plot of Weekly Sentiment vs Stock Price (July – December 2024)

4.4.1 Overall Correlation

The Pearson correlation analysis resulted in a correlation coefficient (r) of +0.0326 with a corresponding p-value of 0.9118. Since the p-value exceeds 0.05 threshold, the result is considered statistically insignificant. Consequently, the null hypothesis cannot be rejected indicating the absence of a meaningful linear association between both the variables. This low r value and high p-value indicated that there is no significant linear relation between sentiment and stock price during the period of the chosen event. This can also suggest that there can be other external factors that can affect the sentiment–market response which cannot be captured using only the social media sentiment data.

Figure 4.5, the scatter plot, confirms this as the data points appear to be widely spread with no trend or clustering. There is no visible pattern or alignment which would suggest correlation. This random scattering can indicate that the graph lacks a clear directional movement. This supports the statistical result that customer sentiment on Reddit and the stock price do not align in a measurable linear fashion during this time frame.

4.4.2 Speculation Phase Correlation

The correlation analysis gives a correlation coefficient (r) of -0.2374 and a p-value of 0.8474. The negative r value suggests a weak inverse relationship where the sentiment score slightly increased, and the stock price tended to decrease. However, the high p-value shows that the relationship is not statistically significant (p-value exceeds 0.05). Consequently, the null hypothesis cannot be rejected implying no meaningful linear relation between both the variables within this period.

While considering the scatter plot in Figure 4.5, it was found that there is a tighter grouping of points compared with the overall period. The downward sloping alignment of some points can be interpreted as a potential inverse visual trend even though the correlation is weak. This can be taken as during the Speculation Phase; the investor behaviour may have been more sensitive to public sentiment due to media influence.

Furthermore, when Figure 4.4, the line-bar chart for sentiment and stock price over time, is considered, there were instances of parallel movement, especially during September. For instance, there is a dip in early September for both sentiment as well as the stock price. This recurring shift can suggest a temporary alignment, showing that the sentiment and stock price react in the same timeframe even if the relationship is non-linear to get a significant correlation coefficient value. This highlights the integration of quantitative and qualitative insights when interpreting the relationship between public sentiment and reputation of a company.

CHAPTER 5: CONCLUSION AND RECOMMENDATIONS

5.1 CONCLUSION

This analysis is to understand the connection between the customer sentiment expressed on a social media platform and Volkswagen's corporate reputation, using stock price as a proxy indicator. The aim was on analysing public reactions surrounding the 2024 factory shutdown and layoff announcement in Germany with the help of Reddit data for sentiment analysis and Yahoo Finance for stock data. This allowed for a real time understanding of public emotional reactions during major corporate events and decisions.

The research focused on a quantitative structure where the timeline of the event from July 2024 till December 2024 which was broken into four distinct periods: Others, Speculation, Negotiation, and Post Agreement. This segmentation helped in understanding the shift in public emotions more precisely on different stages of the event timeline.

VADER sentiment analysis model and correlation testing helped in obtaining more insights from the data that were retrieved and to reach the following conclusions:

1. Public Opinion Breakdown

The pie chart showed that out of 1,533 cleaned comments, around 42.01% comments were positive in nature, 35.16% were negative, and 22.83% were neutral. This indicates that there is a mixed sentiment in the market and there is a largely optimistic market despite the negative context of the factory closure. This reflect the complex nature of the stakeholder perception which should be considered by the company for making a strategic decision.

2. Emotional Impact of Key Reddit Posts

A single Reddit post appeared to be as the most commented post in all the sentiment categories. This suggests that a widely shared post can generate a diverse emotion depending on different context. The popularity of the post highlight how social platform has a significant impact on public opinion and the user engagement.

3. Phase wise Dynamics of Public Sentiment

The weekly sentiment trends, segmented by event phases, reveal fluctuating sentiment level in different event stages. The Speculation Phase showed neutral to slightly positive comments while the Negotiation Phase showed large sentiment fluctuations, whereas a strong sentiment rebound was seen during the Post Agreement Phase. This may indicate the adaptation of the public to the finalized decision.

4. Alignment of Sentiment and Stock Value

The combined sentiment-stock price chart showed some degree of alignment particularly during the Speculation Phase. There was a parallel dip in both public sentiment and the stock value even though the correlation was not statistically significant.

5. Analytical Testing of Sentiment – Market Response

The result of the Pearson Correlation across the entire timeline indicated no statistically significant linear relationship between the sentiment score and the stock price values. However, the scatter plot and the visual analysis showed temporary alignment and behavioural similarities which were only noticed through the visualizations. For instance, in the Speculation Phase, there is no a strict linear or predictive way, but a broader behavioural pattern where both sentiment and the stock price showed dips and recoveries within similar timeframes. This suggests that the sentiment and the stock movement may be related in a nonlinear way which may not be visible through a linear correlation test.

5.2 LIMITATIONS

- The study focuses solely on data collected from a single social media platform, Reddit, which may limit the diversity of customer sentiment and public opinion.
- Although the data was collected till February 2025, missing data entries for early 2025 led to reducing the dataset till December 2024. This limitation may exclude the public reactions that emerged Post Agreement Phase.
- The VADER sentiment analyser used for analysis is mainly suitable for social media short, informal text. However, the inability to detect the context regarding sarcasm, complex expression of mixed sentiment could result in oversimplification of the user opinions. This can thus limit the sentiment interpretation.
- Stock price, which is used as a proxy for corporate reputation, only reflects investor confidence and financial valuation of the company and not the full spectrum of the reputational metrices. This does not account for the range of reputational factors which can include market trends, customer loyalty, and brand equity as well.
- The usage of Pearson Correlation Analysis provides insights to the possible linear alignment between sentiment score and the stock price over the stipulated time but does not show direct influence or causation of these observations.

5.3 RECOMMENDATIONS

To enhance future research, it is advisable to expand the scope to include multiple media platforms such as X, or other relevant platforms, and to integrate news from digital platforms for deeper analysis. Applying advanced sentiment analysis tools could improve the interpretation of sentiment. This can help in capturing insights from a diverse viewpoint and improve the contextual reliability of the results that can be obtained from the analysis.

Recent technological advancements have resulted in the development of more technically advanced tools for sentiment analysis based on machine learning models such as BERT, which can enhance the interpretation of the emotions and have more understanding of the tone of complex texts. These models are capable of detecting nuances such as sarcasm and mixed sentiments, which can often be misclassifies by tools like VADER.

Additionally, using stock price as the sole proxy for the company's reputation reflect only a limited. Therefore, incorporating alternative indicators like brand trust indices could help in building a robust framework to evaluate reputational aspect.

Finally, like the event phase wise segregation, demographic segregation could reveal the differentiated emotional responses. This may offer deeper insights into how various stakeholder groups respond, supporting more strategic decisions and plans for each segment.

5.4 FUTURE WORKS

From the boundaries of this research, future research could extend the methodological and the analytical framework to generate deeper insights into the relationship between social media sentiment and corporate reputation.

(a) Inclusion of multiple data sources

Future research can integrate data from multiple sources where each source can attract different user groups and styles. This can allow the creation of a more meaningful and richer dataset. Combining texts, video transcripts, or image-based reactions can create a cross-modal analysis of public opinion. Comparison of cross-platform studies can also help in determining whether the sentiment dynamics vary according to audience type or the platform.

(b) Exploring event-based and time series

While this research focused on a single high-impact event and its influence on customers, future investigations can include collecting data from several years to capture the recurring sentiment patterns across multiple corporate decisions made by a company. Comparing diverse events like mergers, product recalls, sustainability announcements, and layoffs would help in identifying whether a particular event category evokes reputational shifts.

(c) Adoption of advanced and hybrid analytical model.

The use of transformer-based models such as BERT, RoBERTa (Liu et al., 2019) can be explored, which can interpret context-dependent meanings and domain-specific expressions more accurately than the rule-based systems. Combination of these with lexicon-based systems can help in building a hybrid model with better precision. This would contribute to methodological advancements in sentiment analysis within corporate communication research.

(d) Integrating topic modelling and network analysis

Combining sentiment analysis with topic modelling techniques such as Latent Dirichlet Allocation (LDA) (Farkhod et al., 2021) or BERTopic (Grootendorst, 2022) can identify dominant discussion themes. Also, Social Network Analysis (SNA) (Sheble et al., 2016) could map influential communities that influence the sentiment trends. Integrating these methods would create a multidimensional view of how opinions form and evolve during a corporate event.

(e) Ethical and cross-cultural consideration

Multilingual datasets can be included that could address the ethical and cross-cultural aspects. The collection and usage of data should always adhere to the privacy and fairness standards in social data mining. Comparative analyses across cultural or linguistic contexts may highlight variations in sentiment expression as well as emotional framing, thus contributing to more inclusive models of reputation analysis (Pontiki et al., 2016)

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APPENDIX A: CODE IMPLEMANTATION

(a) Reddit Data Collection and Sentiment Analysis

1. Import Libraries, set up Reddit API and define subreddits and keywords

```
In [14]: import praw
from datetime import datetime
import pandas as pd

# --- Set up Reddit API connection ---
reddit = praw.Reddit(
    client_id="vY0QFuH9UsSTOYeRtthPaQ",
    client_secret="-gPfenC_b-Iea3RDrgXbvBr1E2MW6Q",
    user_agent="reddit-vw-shutdown-scraper"
)

# --- Subreddits to scan ---
subreddits = [
    "Volkswagen", "cars", "autos", "news", "europe",
    "germany", "electricvehicles", "worldnews", "autonews", "AskEurope"
]

# --- Keywords related to factory shutdown and layoffs ---
keywords = [
    "Volkswagen factory shutdown",
    "VW factory closure",
    "Audi Brussels shutdown",
    "Volkswagen layoffs",
    "VW layoffs",
    "Volkswagen job cuts",
    "Audi plant closed",
    "VW production halt",
    "VW shutdown",
    "Audi factory shut down",
    "Volkswagen crisis",
    "Audi Belgium cuts",
    "auto industry layoffs",
    "EV factory slowdown",
    "EV demand drop Volkswagen",
    "Volkswagen to close factory",
    "Volkswagen shuts Brussels plant",
    "Volkswagen Brussels to be shut",
    "Volkswagen halts production",
    "Volkswagen boss wants to close European factories"
]
```

2. Scrape the Reddit posts by keywords and time frame and create data frame

```python
# --- Time settings (for filtering later if needed) ---
start_year = 2024
end_year = 2025

# --- Scrape posts ---
posts_data = []

for sub in subreddits:
    subreddit = reddit.subreddit(sub)
    for keyword in keywords:
        for post in subreddit.search(keyword, sort="new", time_filter="year", limit=200):
            post_date = datetime.fromtimestamp(post.created_utc)
            if post_date.year in [start_year, end_year]:
                posts_data.append({
                    "title": post.title,
                    "text": post.selftext,
                    "created": post_date,
                    "subreddit": sub,
                    "url": post.url,
                    "id": post.id
                })

# --- Create DataFrame ---
df_reddit = pd.DataFrame(posts_data)

print(f"✅ Total relevant posts collected: {len(df_reddit)}")
print(df_reddit[["title", "created", "subreddit"]].head(10))
```

```
✅ Total relevant posts collected: 1337
                                                title             created  \
0               Golf7 1.6tdi 2014 chinese system??? 2025-04-03 22:05:42
1                              Does my car have a DSG? 2025-02-14 23:20:18
2  Can you buy a Volkswagen extended warranty aft... 2025-02-06 14:39:31
3         Why is Volkswagen having such a hard time? 2025-02-05 18:02:27
4             How Long From "Car Built" to Delivery? 2025-01-23 16:41:49
5               What does this mean? 47.822-47.842  2025-01-18 16:02:19
6  If you can, get the Currywurst and Ketchup. Th... 2025-01-18 13:33:27
7  Mechanic Advised: go car shopping— advice needed. 2025-01-16 22:18:50
8  The Volkswagen 1600 TL. The fastback version o... 2024-12-30 01:59:00
9                              VW Jetta Se - CarPlay 2024-12-30 00:06:01

    subreddit
0  Volkswagen
1  Volkswagen
2  Volkswagen
3  Volkswagen
4  Volkswagen
5  Volkswagen
6  Volkswagen
7  Volkswagen
8  Volkswagen
9  Volkswagen
```

3. **Filtering the highly relevant posts using strict keyword matching**

```python
In [16]: # Create the full Reddit DataFrame
df_reddit = pd.DataFrame(posts_data)

# ✅ FILTER for highly relevant posts only (insert this here!)
strict_keywords = [
    "factory shutdown", "plant closure", "factory closure",
    "vw shutdown", "audi shutdown", "volkswagen layoffs",
    "job cuts", "production halt", "vw plant closure",
    "volkswagen to close", "brussels factory", "major layoffs"
]
df_relevant = df_reddit[
    df_reddit["title"].str.lower().str.contains('|'.join(strict_keywords))
]
print(f"✅ Highly relevant posts found: {len(df_relevant)}")

# Optional: View a few results
print(df_relevant[["title", "created", "subreddit"]].head(10))
```

```
✅ Highly relevant posts found: 73
                                                 title              created  \
20   Volkswagen’s Factory Shutdowns & Layoffs Signa... 2024-10-28 20:38:08
28   VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES... 2024-09-08 18:40:18
66   VW Workers Propose Pay Cuts to Prevent Plant C... 2024-11-21 12:21:30
83   VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES... 2024-09-08 18:40:18
106  Volkswagen’s Factory Shutdowns & Layoffs Signa... 2024-10-28 20:38:08
108  VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES... 2024-09-08 18:40:18
110  VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES... 2024-09-08 18:40:18
142  Volkswagen to Close “At Least Three” German Pl... 2024-10-28 11:29:42
145  Volkswagen Faces Potential Job Cuts Amid Cost-... 2024-09-20 19:12:18
150  VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES... 2024-09-08 18:40:18

      subreddit
20   Volkswagen
28   Volkswagen
66   Volkswagen
83   Volkswagen
106  Volkswagen
108  Volkswagen
110  Volkswagen
142  Volkswagen
145  Volkswagen
150  Volkswagen
```

4.Comments extracted from filtered Reddit posts and covert to DataFrame

In [17]:
```python
import pandas as pd
from datetime import datetime
import praw

# Initialize Reddit API
reddit = praw.Reddit(
    client_id="vY0QFuH9UsSTOYeRtthPaQ",
    client_secret="-gPfenC_b-Iea3RDrgXbvBr1E2MW6Q",
    user_agent="YOUR_USER_AGENT"
)

# Assuming df_relevant is already available from your filtered post scraping code
# Make sure the 'id' column exists
if "id" not in df_relevant.columns:
    raise ValueError("Post ID column 'id' is missing in df_relevant. Make sure it is included during post collection.

# Step 1: Extract Comments for Relevant Posts
relevant_comments = []

for post_id in df_relevant["id"]:
    submission = reddit.submission(id=post_id)
    submission.comments.replace_more(limit=0)  # Skip "Load more" links
```

```python
    for comment in submission.comments.list():
        relevant_comments.append({
            "post_id": post_id,
            "comment": comment.body,
            "score": comment.score,
            "created": datetime.fromtimestamp(comment.created_utc)
        })

print(f"\n💬 Total comments collected: {len(relevant_comments)}")

# Convert to DataFrame
df_comments = pd.DataFrame(relevant_comments)
print(df_comments.head())
```

```
💬 Total comments collected: 5480
  post_id                                            comment  score  \
0  1fc23oa  All is well with this but the AI generated pic...     43
1  1fc23oa  The media has been poor at communicating the f...     22
2  1fc23oa  I read a lot of reactions like VW is going ban...      6
3  1fc23oa          Media rides high on hype but short on facts      2
4  1fc23oa  This is the company that killed the B8 Euro Pa...      5

              created
0 2024-09-08 18:52:21
1 2024-09-08 19:18:33
2 2024-09-08 22:51:16
3 2024-09-08 19:32:47
4 2024-09-08 19:31:18
```

5. Validate the extracted comments that belong to the filtered relevant posts

```
In [18]: # Check if all comment post_ids match posts from df_relevant
unmatched_ids = df_comments[~df_comments["post_id"].isin(df_relevant["id"])]

if unmatched_ids.empty:
    print("✅ All comments are from filtered relevant posts.")
else:
    print("⚠️ Warning: Some comments are from unrelated posts!")
    print(unmatched_ids.head())
```

```
✅ All comments are from filtered relevant posts.
```

(a) Perform sentiment analysis on comments and export the final dataset to Excel

```
In [20]: !pip install vaderSentiment
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer

# Step 1: Run Sentiment Analysis on Comments
analyzer = SentimentIntensityAnalyzer()
sentiment_scores = df_comments["comment"].apply(analyzer.polarity_scores)
sentiment_df = sentiment_scores.apply(pd.Series)

# Step 2: Add sentiment scores to df_comments
df_comments = pd.concat([df_comments, sentiment_df], axis=1)

# Step 3: Add categorical sentiment label
df_comments["sentiment"] = df_comments["compound"].apply(
    lambda x: "positive" if x >= 0.05 else "negative" if x <= -0.05 else "neutral"
)

# Step 4: Merge post titles from df_reddit (based on 'id' and 'post_id')
df_final = df_comments.merge(
    df_reddit[["id", "title", "subreddit"]],
    left_on="post_id",
    right_on="id",
    how="left"
)

# Step 5: Drop duplicate 'id' column
df_final.drop(columns=["id"], inplace=True)

# Step 6: Save to Excel
output_file = "reddit_comments_sentiment_with_titles.xlsx"
df_final.to_excel(output_file, index=False)

print(f"✅ Sentiment results (with post titles) saved to: {output_file}")
```

```
Collecting vaderSentiment
  Downloading vaderSentiment-3.3.2-py2.py3-none-any.whl.metadata (572 bytes)
Requirement already satisfied: requests in c:\users\devuj\anaconda3\lib\site-packages (from vaderSentiment) (2.31.0)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\devuj\anaconda3\lib\site-packages (from requests->vaderSentiment) (2.0.4)
Requirement already satisfied: idna<4,>=2.5 in c:\users\devuj\anaconda3\lib\site-packages (from requests->vaderSentiment) (3.4)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\devuj\anaconda3\lib\site-packages (from requests->vaderSentiment) (2.0.7)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\devuj\anaconda3\lib\site-packages (from requests->vaderSentiment) (2024.2.2)
Downloading vaderSentiment-3.3.2-py2.py3-none-any.whl (125 kB)
   ---------------------------------------- 0.0/126.0 kB ? eta -:--:--
   --- ------------------------------------ 10.2/126.0 kB ? eta -:--:--
   --- ------------------------------------ 10.2/126.0 kB ? eta -:--:--
   --- ------------------------------------ 10.2/126.0 kB ? eta -:--:--
   ------------------- ------------------- 61.4/126.0 kB 297.7 kB/s eta 0:00:01
   ------------------------- ------------- 81.9/126.0 kB 353.1 kB/s eta 0:00:01
   ------------------------- ------------- 81.9/126.0 kB 353.1 kB/s eta 0:00:01
   ------------------------- ------------- 81.9/126.0 kB 353.1 kB/s eta 0:00:01
   ------------------------- ------------- 81.9/126.0 kB 353.1 kB/s eta 0:00:01
   ------------------------- ------------- 81.9/126.0 kB 353.1 kB/s eta 0:00:01
   ------------------------- ------------- 81.9/126.0 kB 353.1 kB/s eta 0:00:01
   -------------------------------------- 126.0/126.0 kB 239.1 kB/s eta 0:00:00
Installing collected packages: vaderSentiment
Successfully installed vaderSentiment-3.3.2
✅ Sentiment results (with post titles) saved to: reddit_comments_sentiment_with_titles.xlsx
```

(a) Collection of Stock Data from Yahoo Finance

1. Extract and export historical Volkswagen stock data using yfinance library

```
In [8]: import yfinance as yf
        import pandas as pd

        # Define the company ticker symbol (Volkswagen AG - Frankfurt Stock Exchange)
        ticker = "VOW3.DE"

        # Define the time range (based on your Reddit comment data, around factory shutdown period)
        start_date = "2024-08-01"
        end_date = "2025-04-01"

        # Fetch daily Volkswagen stock data for a wider range
        vw_stock = yf.download("VOW3.DE", start="2024-07-01", end="2025-04-15")

        # Display the first few rows
        vw_stock_data.head()

        # Export to Excel
        vw_stock.to_excel("volkswagen_stock_data.xlsx")

        print("✅ Stock data saved to 'volkswagen_stock_data.xlsx'")

[*********************100%***********************]  1 of 1 completed
✅ Stock data saved to 'volkswagen_stock_data.xlsx'
```

2. Merge weekly sentiment scores with stock prices and export to Excel

```
In [8]: import pandas as pd

# === Step 1: Load weekly sentiment data ===
sentiment_path = r"C:\Users\devuj\OneDrive\Desktop\Dissertation\Thesis\Saved files\Cleaned_data.xlsx"
df_sentiment = pd.read_excel(sentiment_path)

# Convert 'created' column to datetime and resample to weekly sentiment
df_sentiment['created'] = pd.to_datetime(df_sentiment['created'])
df_sentiment.set_index('created', inplace=True)
weekly_sentiment = df_sentiment['compound'].resample('W').mean().reset_index()
weekly_sentiment.rename(columns={'compound': 'Weekly Sentiment'}, inplace=True)

# === Step 2: Load stock data (now with correct headers) ===
stock_path = r"C:\Users\devuj\OneDrive\Desktop\Dissertation\Thesis\Saved files\volkswagen_stock_data.xlsx"
df_stock = pd.read_excel(stock_path)  # No skiprows needed now

# Convert date and resample weekly close price
df_stock['Date'] = pd.to_datetime(df_stock['Date'])
df_stock.set_index('Date', inplace=True)
weekly_stock = df_stock['Close'].resample('W').mean().reset_index()
weekly_stock.rename(columns={'Close': 'Weekly Stock Price'}, inplace=True)

# === Step 3: Merge both datasets on date ===
combined_df = pd.merge(weekly_sentiment, weekly_stock, left_on='created', right_on='Date')
combined_df.drop(columns=['Date'], inplace=True)
combined_df.rename(columns={'created': 'Date'}, inplace=True)

# === Step 4: Save merged result ===
output_path = r"C:\Users\devuj\OneDrive\Desktop\Dissertation\Thesis\Saved files\interpolated_sentiment_vs_stock.xlsx"
combined_df.to_excel(output_path, index=False)

print("✅ Final combined file saved to:")
print(output_path)

✅ Final combined file saved to:
C:\Users\devuj\OneDrive\Desktop\Dissertation\Thesis\Saved files\interpolated_sentiment_vs_stock.xlsx
```

(c) Weekly Sentiment Interpolation

1. Resampling and interpolating the weekly sentiment values and exporting to Excel

```
In [9]: import pandas as pd

# Step 1: Load the cleaned sentiment data
sentiment_path = r"C:\Users\devuj\OneDrive\Desktop\Dissertation\Thesis\Saved files\Cleaned_data.xlsx"
df = pd.read_excel(sentiment_path)

# Step 2: Convert 'created' column to datetime
df['created'] = pd.to_datetime(df['created'])

# Step 3: Set datetime as index
df.set_index('created', inplace=True)

# Step 4: Resample to weekly average sentiment
weekly_sentiment = df['compound'].resample('W').mean()

# Step 5: Interpolate missing sentiment values
weekly_sentiment_interpolated = weekly_sentiment.interpolate(method='linear')

# Step 6: Reset index and save to Excel
weekly_sentiment_interpolated = weekly_sentiment_interpolated.reset_index()
weekly_sentiment_interpolated.rename(columns={'compound': 'Weekly Sentiment', 'created': 'Date'}, inplace=True)

# Step 7: Save the interpolated data to Excel
output_path = r"C:\Users\devuj\OneDrive\Desktop\Dissertation\Thesis\Saved files\interpolated_weekly_sentiment.xlsx"
```

```
weekly_sentiment_interpolated.to_excel(output_path, index=False)

print("✅ Interpolated sentiment saved to:")
print(output_path)
```

```
✅ Interpolated sentiment saved to:
C:\Users\devuj\OneDrive\Desktop\Dissertation\Thesis\Saved files\interpolated_weekly_sentiment.xlsx
```

2. Replacing the old sentiment columns in the combined file with the interpolated sentiment

```
In [11]: import pandas as pd

# Step 1: Load interpolated weekly sentiment
sentiment_path = r"C:\Users\devuj\OneDrive\Desktop\Dissertation\Thesis\Saved files\interpolated_weekly_sentiment.xlsx
df_sentiment = pd.read_excel(sentiment_path)
df_sentiment['Date'] = pd.to_datetime(df_sentiment['Date'])

# Step 2: Load the existing sentiment+stock file
combined_path = r"C:\Users\devuj\OneDrive\Desktop\Dissertation\Thesis\Saved files\interpolated_sentiment_vs_stock.xls
df_combined = pd.read_excel(combined_path)
df_combined['Date'] = pd.to_datetime(df_combined['Date'])

# Step 3: Merge using the updated sentiment
# Drop old sentiment column if it exists
if 'Weekly Sentiment' in df_combined.columns:
    df_combined = df_combined.drop(columns=['Weekly Sentiment'])

# Merge updated sentiment
df_updated = pd.merge(df_combined, df_sentiment, on='Date', how='left')

# Step 4: Save back to the same file (overwrite)
df_updated.to_excel(combined_path, index=False)

print("✅ Sentiment column updated in:")
print(combined_path)
```

```
✅ Sentiment column updated in:
C:\Users\devuj\OneDrive\Desktop\Dissertation\Thesis\Saved files\interpolated_sentiment_vs_stock.xlsx
```

(d) Correlation Analysis

```
[10]: import pandas as pd
from scipy.stats import pearsonr

# Load your merged file with weekly sentiment and stock data
file_path = r"C:\Users\devuj\OneDrive\Desktop\Dissertation\Thesis\Saved files\interpolated_sentiment_vs_stock.xlsx"
df = pd.read_excel(file_path)

# Drop rows with missing values (if any)
df = df.dropna(subset=['Weekly Sentiment', 'Weekly Stock Price'])

# Calculate Pearson correlation and p-value
correlation, p_value = pearsonr(df['Weekly Sentiment'], df['Weekly Stock Price'])

# Print results
print("📊 Pearson Correlation Coefficient:", round(correlation, 4))
print("📉 P-value:", round(p_value, 6))

# Hypothesis testing interpretation
alpha = 0.05
if p_value < alpha:
    print("❗ Result: Reject the null hypothesis (significant correlation exists).")
else:
    print("✅ Result: Fail to reject the null hypothesis (no significant correlation).")
```

```
📊 Pearson Correlation Coefficient: 0.1299
📉 P-value: 0.536151
✅ Result: Fail to reject the null hypothesis (no significant correlation).
```

(e) Visualizations

1. Generating line plot for weekly sentiment and VW stock price

```
In [7]: import pandas as pd
        import yfinance as yf

        # Load saved sentiment data (simulating here)
        sentiment_data = {
            "created": pd.date_range(start="2024-07-01", periods=100, freq="D"),
            "compound": pd.Series(range(100)).apply(lambda x: (x % 10 - 5) / 5.0)
        }
        df_sentiment = pd.DataFrame(sentiment_data)
        df_sentiment['created'] = pd.to_datetime(df_sentiment['created'])
        df_sentiment.set_index('created', inplace=True)

        # Resample sentiment to weekly averages
        weekly_sentiment = df_sentiment['compound'].resample('W').mean()

        # Fetch stock data
        vw_stock = yf.download("VOW3.DE", start="2024-07-01", end="2025-04-15")
```

```
weekly_stock = vw_stock['Close'].resample('W').mean()

# Align sentiment index with stock and interpolate missing weeks
weekly_sentiment_aligned = weekly_sentiment.reindex(weekly_stock.index)
weekly_sentiment_aligned.interpolate(method='time', inplace=True)

# Combine and save
combined_df = pd.concat([weekly_sentiment_aligned, weekly_stock], axis=1)
combined_df.columns = ['Weekly Sentiment', 'Weekly Stock Price']
combined_df.dropna(inplace=True)

print("✅ Displaying top rows of the combined dataset:")
print(combined_df.head(10))

# Optionally: visualize directly using matplotlib
import matplotlib.pyplot as plt

combined_df.plot(figsize=(12, 6), title="Weekly Sentiment vs VW Stock Price", grid=True)
plt.xlabel("Week")
plt.ylabel("Value")
plt.tight_layout()
plt.show()

# Save to Excel
combined_df.to_excel("interpolated_sentiment_vs_stock.xlsx", index=True)
```

```
[*********************100%***********************]  1 of 1 completed
```

```
✅ Displaying top rows of the combined dataset:
            Weekly Sentiment  Weekly Stock Price
Date
2024-07-07         -0.400000          106.550000
2024-07-14         -0.142857          107.070001
2024-07-21          0.114286          106.959998
2024-07-28         -0.200000          105.420001
2024-08-04         -0.228571          101.518001
2024-08-11          0.028571           93.896002
2024-08-18          0.000000           93.800002
2024-08-25         -0.314286           96.472000
2024-09-01         -0.057143           96.444000
2024-09-08          0.200000           95.087999
```

2. Identifying the most discussed posts in each sentiment category

```
[9]: import pandas as pd

# Load the Excel file
df = pd.read_excel(r"C:\Users\devuj\OneDrive\Desktop\Dissertation\Thesis\Saved files\Cleaned_data.xlsx")

# Group by 'title' and 'sentiment' to count the number of comments per post per sentiment
sentiment_counts = df.groupby(['title', 'sentiment']).size().reset_index(name='Comment_Count')

# Get the top 1 post for each sentiment category
top_positive = sentiment_counts[sentiment_counts['sentiment'] == 'positive'].sort_values(by='Comment_Count', ascending=False).head(1)
top_negative = sentiment_counts[sentiment_counts['sentiment'] == 'negative'].sort_values(by='Comment_Count', ascending=False).head(1)
top_neutral = sentiment_counts[sentiment_counts['sentiment'] == 'neutral'].sort_values(by='Comment_Count', ascending=False).head(1)

# Combine the results
top_combined = pd.concat([top_positive, top_negative, top_neutral])

# Display the result in Jupyter notebook
top_combined
```

[9]:

	title	sentiment	Comment_Count
6	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES...	positive	3375
4	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES...	negative	4500
5	VOLKSWAGEN FACES TOUGH CHOICES: PLANT CLOSURES...	neutral	2925

3. Generating the sentiment distribution table

```
In [12]: import pandas as pd

# Load the cleaned sentiment data
file_path = r"C:\Users\devuj\OneDrive\Desktop\Dissertation\Thesis\Saved files\Cleaned_data.xlsx"
df = pd.read_excel(file_path)

# Ensure the sentiment column is named 'sentiment'
# If it's named differently, update the column name accordingly
if 'sentiment' not in df.columns:
    print("❗ Column 'sentiment' not found. Please check the column names:")
    print(df.columns)
else:
    # Get sentiment distribution
    sentiment_counts = df['sentiment'].value_counts().reset_index()
    sentiment_counts.columns = ['Sentiment', 'Count']

    print("📊 Sentiment Distribution Table:")
    print(sentiment_counts)
```

```
📊 Sentiment Distribution Table:
  Sentiment  Count
0  negative   7390
1  positive   6419
2   neutral   4816
```

APPENDIX B: PROPOSAL FORM

Dissertation Proposal Form

Student Name	Devika Krishnan	**Supervisor Name**	Stefan Lutz
Student Number	STU199943	**Course**	MSc Data Analytics and Information Systems Management
Project title:	Using Sentiment Analysis to Analyze Company Reputation		
Proposed activity – aims, objectives, research question(s), and state how it is novel			

Aim

The main aim of this project is to use sentiment analysis to analyse company reputation (considering Volkswagen).

Research Objective

- To analyse customer sentiment around the aforementioned company events using Reddit or Twitter data.
- To assess whether customer sentiment on social media can affect the company's reputation.
- To visualize sentiment trends using appropriate data visualization techniques.
- To correlate the sentiment analysis with company's stock price as an indicator of corporate reputation.

Research Questions

RQ 1: How does customer sentiment change during key events for the company?

RQ 2: Can social media sentiment influence company's reputation, as reflected by stock prices?

RQ 3: What visual insights can be derived from the analysis of sentiment trends?

Research Novelty

Online chatter or user-generated content is a powerful tool used for forecasting stock market performance. It can highlight both negative and positive sentiments, which serve as strong indicators of market outcomes (Tirunillai and Tellis., 2011). In the era of social media, with increasing user engagement, the sentiments expressed by users can be an asset for analyzing a company's impact on society. During events like a new product release or a significant company change, customer sentiment is affected, and this can be analyzed through sentiment analysis, also known as opinion mining.

Sentiment analysis is the study of people's opinions, emotions, and attitudes as expressed in written text. It is a key area within Natural Language Processing (NLP), expanding into data mining and web mining to help understand public sentiment (Liu, 2012).

This study focuses on the application of real-time sentiment analysis to assess the relationship between customer emotions and a company's reputation. Although there has been prior research in this area, the focus here is to gain insights into how public perceptions influence stock prices, which can be used as a factor representing a company's reputation.

Methodology – rationale, data selection and collection, recruitment, participant demographics, analytical process

Rationale

In this new digital era, social media plays a crucial role in shaping a company's reputation by providing a platform where customers express their opinions and experiences about products and services. Sentiment analysis focuses on determining the sentiment behind these opinions, whether they are positive, negative, or neutral (Classification of Customer Reviews Based on Sentiment Analysis). This analysis can help businesses understand customer feedback, predict market trends, and manage public relations effectively.

This research focuses on exploring customer sentiment towards Volkswagen using Reddit data. These platforms generate high-volume, real-time data that can help analyze customer emotions in response to new changes introduced by the company in its products and services. Using data from these social media sources is advantageous over traditional customer surveys, which are often time-consuming. Social media data is more spontaneous and provides opinions from a diverse audience.

By analyzing tweets or posts related to the company, this research aims to understand shifts in customer sentiment and how these shifts correlate with the company's stock price, which, in turn, reflects its overall reputation. The findings from this research can aid in making informed decisions and understanding the preferences of the target market, helping companies develop better marketing and product strategies. Sentiment analysis can also help companies build trust with the public, thereby improving their performance in a changing market environment.

Data Selection and Collection

The data will be collected from social media platforms like Twitter or Reddit through their respective APIs. For Twitter, the Tweepy library will be used to gather tweets that mention the company, using company-related hashtags to collect real-time data around specific events (Pak and Paroubek, 2009). Similarly, if Reddit is chosen as the platform, tools like BeautifulSoup or the Python Reddit API Wrapper (PRAW) will be used to extract relevant text data. The collected data should focus on posts or discussions that pertain to current events or company-related topics.

Once the data is collected, it will be cleaned by removing noise such as special characters, URLs, and other irrelevant elements. The text will then be tokenized to break it into individual words, with stopwords being removed to make the dataset suitable for analysis (Gräbner et al., 2012).

As for the stock price data, it will be sourced from reliable platforms such as Google Finance using the yfinance Python library (Tirunillai and Tellis, 2011).

Recruitment / Participant Demographics

This study relies on secondary data that are collected from publicly available sources like Twitter or Reddit.

Analytical Process

This study involves several steps, focusing on **sentiment analysis**, **visualization**, and **correlation analysis**.

1. **Sentiment Analysis**: After data collection and cleaning, the dataset will be analyzed using the Valence Aware Dictionary and sEntiment Reasoner (VADER), a lexicon-based sentiment analysis tool. VADER is designed specifically for social media and short text analysis, where it classifies the text into positive, negative, and neutral sentiments, or provides a compound score representing the overall sentiment (Hutto and Gilbert, 2014).

2. **Visualization**: Once the sentiment scores are obtained, data visualization techniques will be used to enhance understanding of the dataset and sentiment trends. Matplotlib and Seaborn are libraries that can assist with this. For example, pie charts can display the distribution of sentiments, line graphs can illustrate sentiment changes over time, and bar charts can show the volume of sentiment classifications over specific periods. The choice of graphs will depend on the specific objectives we aim to achieve.

3. **Correlation Analysis**: Correlation analysis, as an inferential statistical method, will be used to explore the relationship between sentiment analysis results and stock prices. This analysis will help determine whether there is a significant relationship between customer sentiment and changes in the company's stock performance, which serves as an indicator of the company's reputation (Tirunillai and Tellis, 2011).

References

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Project management

Table: Project timeline and key outputs

Date and Month	**Activity**
September	Project Planning
November	Literature Review and data selection
December	Data Preprocessing
January	Sentiment Analysis and Visualization
February	Correlation Analysis and Conclusion

Supervision Meetings

Research Data Management Plan (Describe the data you expect to acquire or generate during this research project, how you will manage, describe, analyse, and store the data and what mechanisms you will use to share and preserve your data.)

Data Collection

The data source for this research includes:

- Social Media Posts: Data will be collected from Twitter or Reddit to gather real time opinion about the company. Relevant tweets or posts including the discussion about the company will be extracted using Twitter API (with Tweepy library) or Reddit API (with PRAW).
- Stock Price Data: Historic stock price of the company will be collected from reliable financial platform like Google Finance using yfinance library.

Data Management

The collected data will be accessible only to the research team to ensure data integrity and

security. To protect users' privacy, all personally identifiable information will be excluded.
Planned outputs/publications/research datasets/impact/dissemination
The planned output of this research will be a detailed report that includes the methodology, findings, and implications of sentiment analysis on company reputation. The findings can contribute to the fields of data science, business analytics, and related areas, allowing researchers to extend studies on the relationship between sentiment analysis and company reputation as part of future work. Based on the findings, recommendations can be formulated to help business professionals better understand reputation management and develop strategies to improve customer engagement. This will enhance the impact of the study by sharing insights into how real-time sentiment analysis can aid in understanding and managing company reputation.

ARDEN UNIVERSITY

APPENDIX C: ETHICS APPROVAL

Certificate of Ethical Approval

Applicant:	Ms. Devika Krishnan (STU199943)
Project Title:	Using Sentiment Analysis to Analyze Company Reputation

This is to certify that the above named applicant has completed the Arden Research Management System Ethical Approval process and their project has been confirmed and approved as Low Risk

Date of approval:	30 Dec 2024
Project Reference Number:	P10638